

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 22 — Analytical Chemistry: IR & Mass Spectrometry (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	22.1 Infrared Spectroscopy: Principles, Absorption Bands & Greenhouse Effect · 22.2 Mass Spectrometry: Molecular Ions, [M+], [M+2] & Fragmentation
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

- 22.1 Infrared Spectroscopy: Principles, Bands & Greenhouse Effect** Infrared spectroscopy (molecular vibrations, stretching and bending, requirement for changing dipole moment)
- 22.2 Mass Spectrometry: Molecular Ions, [M+], [M+2] & Fragmentation** Mass spectrometry instrumentation (electron impact ionization, acceleration, magnetic sector deflection by m/z)
- High-Frequency Core Repeats** — The 10 most frequently examined Cambridge Paper 1 questions on Analytical Chemistry (Q101 – Q110).

SUBTOPIC 22.1 — INFRARED SPECTROSCOPY: PRINCIPLES, BANDS & GREENHOUSE EFFECT (Q1 – Q55)

1 [HARD] Distinguishing Alcohol from Carboxylic Acid by IR — 9701/11/M/J/23/Q38

[1]

Two unknown organic liquids, X and Y, both contain oxygen. The infrared spectrum of Compound X shows a smooth, broad absorption band centered at 3350 cm^{-1} and NO absorption in the $1650\text{--}1750\text{ cm}^{-1}$ region. The infrared spectrum of Compound Y shows a very broad, ragged absorption band extending from 2500 to 3300 cm^{-1} overlapping C-H stretches, alongside an intense, sharp peak at 1715 cm^{-1} .

What functional groups are present in Compound X and Compound Y?

- [A] Compound X contains an ester group; Compound Y contains an alcohol group.
- [B] Compound X contains an alcohol group ($-\text{OH}$); Compound Y contains a carboxylic acid group ($-\text{COOH}$).
- [C] Compound X contains a primary amine; Compound Y contains an aldehyde.
- [D] Compound X contains a ketone; Compound Y contains an ether.

2 [EASY] Greenhouse Effect: Molecular Dipole Requirement — 9701/12/M/J/23/Q38

[1]

Why do water vapor (H_2O), carbon dioxide (CO_2), and methane (CH_4) act as greenhouse gases by absorbing infrared radiation emitted by the Earth's surface, whereas nitrogen gas (N_2) and oxygen gas (O_2) do not?

- [A] N_2 and O_2 are completely transparent to all electromagnetic radiation.
- [B] Methane and carbon dioxide are liquids in the upper atmosphere.
- [C] Greenhouse gases have higher relative molecular masses than oxygen.
- [D] Bending and asymmetric stretching vibrations in H_2O , CO_2 , and CH_4 produce a change in molecular dipole moment, which allows them to absorb infrared photons, whereas homonuclear diatomic molecules (N_2 and O_2) have zero dipole moment in all vibrational states.

3 [MEDIUM] Differentiating Aldehyde, Ketone, and Ester by IR — 9701/13/M/J/23/Q38

[1]

An unknown organic liquid has a molecular formula of $\text{C}_3\text{H}_6\text{O}_2$. Its infrared spectrum shows an intense, sharp absorption at 1740 cm^{-1} , strong absorption bands at $1150\text{--}1250\text{ cm}^{-1}$, and NO absorption between 2500 and 3600 cm^{-1} .

What is the class of compound?

- [A] An ester (such as ethyl methanoate or methyl ethanoate)
- [B] A carboxylic acid (propanoic acid)
- [C] A hydroxy ketone
- [D] An unsaturated dialcohol

4 [EASY] Infrared Fingerprint Region ($< 1500\text{ cm}^{-1}$) — 9701/11/O/N/23/Q38

[1]

What is the significance of the 'fingerprint region' (wavenumbers below 1500 cm^{-1}) in an infrared spectrum?

- [A] It shows only absorption by single carbon-carbon bonds and is identical for all organic molecules.
- [B] It measures the exact boiling point of the substance.
- [C] It contains a highly complex pattern of bending and skeletal vibrations unique to a specific molecule, allowing definitive identification by direct computer matching against a spectral database of known compounds.
- [D] It detects impurities by turning the spectrum green.

5 [EASY] Nitrile Absorption Band — 9701/13/M/J/22/Q5

[1]

Which characteristic infrared absorption band confirms the presence of a nitrile group ($-\text{C}\equiv\text{N}$) in an organic compound?

- [A] A sharp, distinct peak at $2220\text{--}2260\text{ cm}^{-1}$
- [B] A broad band at 3300 cm^{-1}
- [C] A sharp peak at 1715 cm^{-1}
- [D] A doublet at 1450 cm^{-1}

6 [HARD] Primary vs Secondary Amine N-H Bands — 9701/11/M/J/221/Q6**[1]**

How does the infrared spectrum of a primary aliphatic amine (e.g. ethylamine, $\text{CH}_3\text{CH}_2\text{NH}_2$) differ from that of a secondary amine (e.g. dimethylamine, $(\text{CH}_3)_2\text{NH}$)?

- [A] A primary amine has no peaks above 3000 cm^{-1} .
- [B] A secondary amine shows an intense carbonyl peak at 1710 cm^{-1} .
- [C] A primary amine shows a broad band from 2500 to 3000 cm^{-1} .
- [D] A primary amine exhibits two distinct N-H stretching spikes (a doublet) between 3300 and 3500 cm^{-1} , whereas a secondary amine exhibits only a single N-H spike.

7 [EASY] Alkene C=C vs C-H Stretching — 9701/12/M/J/222/Q7**[1]**

In the infrared spectrum of an alkene (e.g. hex-1-ene), what absorption bands confirm the presence of the C=C double bond?

- [A] A broad band at 3400 cm^{-1} .
- [B] A sharp C=C stretch at $1620\text{--}1680\text{ cm}^{-1}$ and an unsaturated =C-H stretch just above 3000 cm^{-1} ($3010\text{--}3095\text{ cm}^{-1}$).
- [C] A strong peak at 1720 cm^{-1} .
- [D] A peak at 2250 cm^{-1} .

8 [HARD] Symmetrical Alkenes: IR Inactivity — 9701/13/M/J/223/Q8**[1]**

Why does trans-hex-3-ene show NO absorption band for its C=C double bond in its infrared spectrum, whereas hex-1-ene shows a clear C=C band at 1650 cm^{-1} ?

- [A] trans-hex-3-ene has no double bond.
- [B] trans-hex-3-ene is an ionic solid.
- [C] trans-hex-3-ene is centrosymmetric; stretching of the central C=C bond produces zero change in molecular dipole moment, making it IR-inactive.
- [D] hex-1-ene contains an oxygen atom.

9 [EASY] Aldehyde C-H Fermi Resonance Doublet — 9701/11/M/J/224/Q9**[1]**

In addition to the C=O stretch at 1725 cm^{-1} , what distinctive C-H absorption bands identify an aldehyde (R-CHO)?

- [A] A broad band at 3400 cm^{-1} .
- [B] A characteristic pair of weak-to-medium C-H stretching spikes at 2720 cm^{-1} and 2820 cm^{-1} (the aldehyde Fermi resonance doublet).
- [C] A single spike at 2250 cm^{-1} .
- [D] A peak at 1650 cm^{-1} .

10 [HARD] Monitoring Oxidation of Alcohol by IR — 9701/12/M/J/220/Q10**[1]**

When propan-2-ol is oxidized to propanone by heating with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, how does the infrared spectrum of the reaction mixture change?

- [A] The broad O-H absorption band at 3350 cm^{-1} disappears, and an intense, sharp carbonyl (C=O) peak appears at 1715 cm^{-1} .
- [B] A nitrile band appears at 2250 cm^{-1} .
- [C] The C-H stretch peaks vanish completely.
- [D] An alkene peak appears at 1650 cm^{-1} .

11 [EASY] Monitoring Esterification by IR — 9701/13/M/J/221/Q11**[1]**

When ethanoic acid and ethanol react to form ethyl ethanoate, how can infrared spectroscopy confirm that the esterification was successful?

- [A] A triple bond peak appears at 2250 cm^{-1} .
- [B] All C-H peaks disappear.
- [C] The entire spectrum becomes a flat line.
- [D] The very broad carboxylic acid O-H band ($2500\text{--}3300\text{ cm}^{-1}$) and alcohol O-H band (3350 cm^{-1}) disappear, leaving a sharp ester carbonyl peak at $1735\text{--}1750\text{ cm}^{-1}$ and strong C-O single bond stretches at 1240 cm^{-1} .

12 [HARD] Wavenumber Units (cm^{-1}) Meaning — 9701/11/M/J/222/Q12**[1]**

What physical quantity does the unit wavenumber (cm^{-1}) represent in infrared spectroscopy?

- [A] The speed of the molecule in centimeters per second.
- [B] The thickness of the liquid sample in centimeters.
- [C] The number of waves per centimeter (wavenumber = $1 / \text{wavelength}$), which is directly proportional to the vibrational frequency and energy of the absorbed photon.
- [D] The mass of the sample in grams per centimeter.

13 [EASY] Factors Determining Bond Vibrational Frequency — 9701/12/M/J/223/Q13**[1]**

According to Hooke's law, why does a C=O double bond vibrate at a higher frequency ($\sim 1715 \text{ cm}^{-1}$) than a C-O single bond ($\sim 1100 \text{ cm}^{-1}$)?

- [A] A double bond has a higher bond order and greater force constant (stiffness), requiring more energy to stretch than a weaker single bond.
- [B] Oxygen has a higher mass in a double bond.
- [C] Double bonds are longer than single bonds.
- [D] Single bonds contain more electrons.

14 [HARD] Effect of Atomic Mass on Vibrational Frequency — 9701/13/M/J/224/Q14**[1]**

Why does a C-H bond vibrate at a much higher frequency ($\sim 2950 \text{ cm}^{-1}$) than a C-Cl bond ($\sim 700 \text{ cm}^{-1}$)?

- [A] C-Cl bonds are much stronger than C-H bonds.
- [B] Chlorine atoms have zero mass.
- [C] Hydrogen has an extremely small atomic mass (1.0) compared to chlorine (35.5), resulting in a much smaller reduced mass and a vastly higher vibrational frequency.
- [D] Hydrogen forms double bonds with carbon.

15 [EASY] Conjugation Effect on Carbonyl Absorption — 9701/11/M/J/220/Q15**[1]**

Why does the carbonyl stretch in propenal ($\text{CH}_2=\text{CH}-\text{CHO}$, 1685 cm^{-1}) occur at a LOWER wavenumber than in saturated propanal ($\text{CH}_3\text{CH}_2\text{CHO}$, 1730 cm^{-1})?

- [A] Propenal contains no double bonds.
- [B] Delocalization of pi electrons between the C=C and C=O bonds gives the carbonyl bond partial single-bond character, reducing its force constant and lowering the vibrational frequency.
- [C] Conjugation makes the carbonyl bond triple.
- [D] Propenal has a higher molecular mass.

16 [HARD] Hydrogen Bonding Effect on O-H Frequency — 9701/12/M/J/221/Q16**[1]**

Why does an alcohol in dilute gas phase show a sharp O-H peak at $\sim 3650 \text{ cm}^{-1}$, whereas liquid ethanol shows a broad band at $\sim 3350 \text{ cm}^{-1}$?

- [A] In the gas phase, the oxygen atom falls off.
- [B] Liquid ethanol is an ionic liquid.
- [C] Hydrogen bonding increases the force constant of the O-H bond.
- [D] In the liquid phase, extensive intermolecular hydrogen bonding weakens and lengthens the covalent O-H bonds, lowering their force constants and broadening the range of vibrational frequencies.

17 [EASY] Infrared Spectrum of Carboxylic Acid Anhydride — 9701/13/M/J/222/Q17**[1]**

What unique spectral pattern identifies a carboxylic acid anhydride (e.g. ethanoic anhydride, $(\text{CH}_3\text{CO})_2\text{O}$)?

- [A] A single broad band at 3300 cm^{-1} .
- [B] A sharp triple bond peak at 2150 cm^{-1} .
- [C] Two distinct carbonyl stretching bands separated by $\sim 60 \text{ cm}^{-1}$ (at $\sim 1760 \text{ cm}^{-1}$ and $\sim 1820 \text{ cm}^{-1}$) due to coupled symmetric and asymmetric vibrations of the two carbonyl groups.
- [D] No carbonyl peaks above 1600 cm^{-1} .

18 [HARD] Acyl Chloride Carbonyl Frequency — 9701/11/M/J/223/Q18**[1]**

Why does ethanoyl chloride (CH_3COCl) exhibit its carbonyl stretch at an exceptionally HIGH wavenumber ($\sim 1800\text{ cm}^{-1}$)?

- [A] The strongly electronegative chlorine atom withdraws electron density inductively from the carbonyl carbon ($-I$ effect), strengthening the $\text{C}=\text{O}$ bond and increasing its force constant.
- [B] Chlorine donates electrons by resonance.
- [C] The $\text{C}=\text{O}$ bond becomes a single bond.
- [D] Chlorine forms hydrogen bonds.

19 [EASY] Amide Carbonyl (Amide I) Frequency — 9701/12/M/J/224/Q19**[1]**

Why does the carbonyl stretch in ethanamide (CH_3CONH_2 , the 'Amide I' band) occur at a relatively LOW wavenumber ($\sim 1650\text{--}1680\text{ cm}^{-1}$)?

- [A] Ethanamide contains no carbonyl group.
- [B] Nitrogen is more electronegative than fluorine.
- [C] Ethanamide forms ionic bonds with nitrogen.
- [D] Strong resonance donation of the nitrogen lone pair into the carbonyl group gives the $\text{C}=\text{O}$ bond substantial single-bond character, lowering its force constant.

20 [HARD] Infrared Spectrum of Arenes (Benzene Ring) — 9701/13/M/J/220/Q20**[1]**

What combination of absorption bands confirms the presence of an aromatic benzene ring in an unknown compound?

- [A] A single broad band at 3400 cm^{-1} .
- [B] Weak $\text{sp}^2\text{ C-H}$ stretches just above 3000 cm^{-1} ($3030\text{--}3100\text{ cm}^{-1}$), sharp aromatic $\text{C}=\text{C}$ ring breathing absorptions at $1450\text{--}1600\text{ cm}^{-1}$, and strong C-H out-of-plane bending bands below 900 cm^{-1} .
- [C] A sharp peak at 2250 cm^{-1} .
- [D] An intense ester band at 1740 cm^{-1} .

21 [EASY] Infrared Inactivity of Nitrogen and Oxygen Molecules — 9701/11/M/J/221/Q21**[1]**

Why are atmospheric nitrogen (N_2) and oxygen (O_2) completely transparent to infrared radiation?

- [A] N_2 and O_2 have no chemical bonds.
- [B] Being non-polar, homonuclear diatomic molecules, their bond stretching vibrations produce zero change in dipole moment, making them completely IR-inactive.
- [C] N_2 and O_2 exist only at absolute zero.
- [D] They absorb only green light.

22 [HARD] Greenhouse Warming Potential: Methane vs CO_2 — 9701/12/M/J/222/Q22**[1]**

Why is methane (CH_4) over 25 times more potent as a greenhouse gas per molecule than carbon dioxide (CO_2) over a 100-year timescale?

- [A] Methane reacts with oxygen to form uranium.
- [B] Methane is a solid in the air.
- [C] Carbon dioxide absorbs zero infrared radiation.
- [D] Methane possesses nine vibrational degrees of freedom and absorbs terrestrial infrared radiation in spectral 'atmospheric window' wavelengths (around 7.7 micrometers / 1300 cm^{-1}) where CO_2 and H_2O do not absorb.

23 [EASY] Water Vapor as Primary Natural Greenhouse Gas — 9701/13/M/J/223/Q23**[1]**

Why is water vapor (H_2O) the single largest contributor to the Earth's natural greenhouse effect?

- [A] Water molecules have an asymmetric bent geometry with a permanent dipole moment, producing intense IR absorption across wide vibrational and rotational bands, and water is abundant in the troposphere.
- [B] Water vapor is non-polar.
- [C] Water vapor absorbs only ultraviolet radiation.
- [D] Water vapor is an artificial pollutant.

24 [HARD] Carbon Monoxide (CO) IR Activity — 9701/11/M/J/224/Q24**[1]**

Is carbon monoxide (CO) an infrared-active gas, and why?

- [A] No, because CO is a diatomic molecule.
- [B] No, because CO contains no hydrogen atoms.
- [C] Yes, because CO has a permanent dipole moment ($\text{C}^{\delta-} - \text{O}^{\delta+}$) and its stretching vibration produces an oscillating dipole moment that absorbs IR radiation at $\sim 2143 \text{ cm}^{-1}$.
- [D] Yes, because CO is an addition polymer.

25 [EASY] Infrared Sampling: ATR (Attenuated Total Reflectance) — 9701/12/M/J/220/Q25**[1]**

What is the primary practical advantage of modern Attenuated Total Reflectance (ATR) sampling in infrared spectroscopy compared to traditional KBr pellet preparation?

- [A] Solid and liquid samples can be analyzed directly in seconds without requiring grinding, solvent dissolution, or pressing into delicate salt discs.
- [B] ATR uses radioactive lasers.
- [C] ATR works only in outer space.
- [D] ATR converts samples into pure water.

26 [HARD] Beer-Lambert Law in Quantitative IR Analysis — 9701/13/M/J/221/Q26**[1]**

According to the Beer-Lambert Law ($A = \epsilon \cdot c \cdot l$), how is the absorbance (A) of an infrared band related to the concentration (c) of a functional group in a solution?

- [A] Absorbance is inversely proportional to concentration.
- [B] Absorbance equals concentration squared.
- [C] Absorbance is completely independent of concentration.
- [D] Absorbance is directly proportional to both concentration and path length: $A = \epsilon \cdot c \cdot l$.

27 [EASY] Distinguishing Alkyne: Terminal vs Internal — 9701/11/M/J/222/Q27**[1]**

How does the infrared spectrum of a terminal alkyne (e.g. pent-1-yne) differ from that of an internal alkyne (e.g. pent-2-yne)?

- [A] Pent-2-yne shows a broad O-H band.
- [B] Pent-1-yne shows a sharp terminal sp C-H stretch at 3300 cm^{-1} and a $\text{C}\equiv\text{C}$ stretch at 2120 cm^{-1} , whereas pent-2-yne lacks the 3300 cm^{-1} peak and has an extremely weak or silent $\text{C}\equiv\text{C}$ band.
- [C] Pent-1-yne shows a carbonyl peak at 1715 cm^{-1} .
- [D] Both have identical spectra.

28 [HARD] Overtone and Combination Bands in IR — 9701/12/M/J/223/Q28**[1]**

What are 'overtones' in an infrared spectrum, and at what wavenumbers do they appear relative to fundamental absorptions?

- [A] Peaks caused by dust on the lens.
- [B] Peaks caused by nuclear magnetic resonance.
- [C] Transitions from ground state to higher vibrational energy levels ($v = 0$ to $v = 2$), appearing at approximately twice the fundamental wavenumber ($2 \cdot \nu_0$) with much lower intensity.
- [D] Peaks that appear at half the fundamental frequency.

29 [EASY] Infrared Spectra of Inorganic Polyatomic Ions — 9701/13/M/J/224/Q29**[1]**

Which inorganic polyatomic ion exhibits a strong, broad absorption band around 1400 cm^{-1} and a sharp peak at 875 cm^{-1} due to asymmetric stretching and out-of-plane bending?

- [A] The chloride ion (Cl^-)
- [B] The carbonate ion (CO_3^{2-})
- [C] The sodium ion (Na^+)
- [D] The sulfide ion (S^{2-})

30 [HARD] Identification of Carboxylic Acid Salt by IR — 9701/11/M/J/220/Q30**[1]**

How does converting propanoic acid into sodium propanoate ($\text{CH}_3\text{CH}_2\text{COONa}$) alter the infrared spectrum in the carbonyl region?

- [A] The sharp carbonyl peak at 1715 cm^{-1} disappears and is replaced by two prominent carboxylate bands (asymmetric stretch at $1550\text{--}1610\text{ cm}^{-1}$ and symmetric stretch at 1400 cm^{-1}).
- [B] The spectrum becomes identical to that of an alkane.
- [C] A broad band appears at 3500 cm^{-1} .
- [D] A triple bond peak appears at 2250 cm^{-1} .

31 [EASY] IR Spectrum of Halogenoalkanes: C–X Region — 9701/12/M/J/221/Q31**[1]**

Why is infrared spectroscopy relatively ineffective for routine identification of C–Cl and C–Br bonds compared to mass spectrometry or NMR?

- [A] C–Cl and C–Br bonds produce zero dipole moments.
- [B] Halogenoalkanes evaporate instantly in the spectrometer.
- [C] Halogens absorb only radio waves.
- [D] C–Cl ($600\text{--}800\text{ cm}^{-1}$) and C–Br ($500\text{--}600\text{ cm}^{-1}$) absorptions occur in the low-wavenumber region where skeletal fingerprint bands overlap and salt prism optics become opaque.

32 [HARD] Deducing Isomer Identity from IR Spectrum — 9701/13/M/J/222/Q32**[1]**

Two isomers have the molecular formula $\text{C}_2\text{H}_6\text{O}$. Isomer A shows a broad absorption band at 3350 cm^{-1} . Isomer B shows NO absorption above 3000 cm^{-1} or in the $1600\text{--}1800\text{ cm}^{-1}$ region, but strong absorption at 1120 cm^{-1} .

What are Isomers A and B?

- [A] Isomer A is ethanoic acid; Isomer B is ethanal.
- [B] Isomer A is ethanal; Isomer B is ethanol.
- [C] Isomer A is ethanol ($\text{CH}_3\text{CH}_2\text{OH}$); Isomer B is methoxymethane (dimethyl ether, $\text{CH}_3\text{--O--CH}_3$).
- [D] Isomer A is ethene; Isomer B is ethane-1,2-diol.

33 [EASY] Deducing Functional Group: Carbonyl at 1715 vs 1740 cm^{-1} — 9701/11/M/J/223/Q33**[1]**

An unknown ester and an unknown ketone are analyzed. Sample 1 has an intense band at 1715 cm^{-1} . Sample 2 has an intense band at 1745 cm^{-1} .

Which sample is the ester?

- [A] Sample 2 is the ester; Sample 1 is the ketone.
- [B] Sample 1 is the ester; Sample 2 is the ketone.
- [C] Both are esters.
- [D] Neither is an ester.

34 [HARD] Infrared Spectrum of Nitroalkanes (--NO_2) — 9701/12/M/J/224/Q34**[1]**

What characteristic absorption bands confirm the presence of an aliphatic nitro group (--NO_2) in an organic molecule?

- [A] A single broad band at 3400 cm^{-1} .
- [B] A sharp peak at 2250 cm^{-1} .
- [C] Two intense, sharp bands due to asymmetric and symmetric N–O stretching at $1530\text{--}1560\text{ cm}^{-1}$ and $1350\text{--}1380\text{ cm}^{-1}$.
- [D] A doublet at $3300\text{--}3500\text{ cm}^{-1}$.

35 [EASY] Infrared Spectrum of Sulfoxides and Sulfones — 9701/13/M/J/220/Q35**[1]**

At what wavenumbers do the sulfur–oxygen stretching vibrations appear in sulfoxides ($\text{R}_2\text{S=O}$) and sulfones (R_2SO_2)?

- [A] Both absorb at 3300 cm^{-1} .
- [B] Sulfoxides show a strong S=O stretch at $1030\text{--}1070\text{ cm}^{-1}$; sulfones show two strong bands at $1120\text{--}1160\text{ cm}^{-1}$ and $1300\text{--}1350\text{ cm}^{-1}$.
- [C] Both absorb at 1715 cm^{-1} .
- [D] Neither absorbs infrared radiation.

36 [HARD] Infrared Spectrum of Polycyclic Aromatic Hydrocarbons — 9701/11/M/J/221/Q36**[1]**

Why do polycyclic aromatic hydrocarbons (PAHs) show complex, intense fingerprint bands in the 700–900 cm^{-1} region?

- [A] They contain sulfur crosslinks.
- [B] They undergo rapid photochemical cleavage.
- [C] They are non-polar alkanes.
- [D] They have multiple adjacent aromatic C-H bonds that undergo coupled out-of-plane (oop) bending vibrations characteristic of their ring fusion geometry.

37 [EASY] Vibrational Degrees of Freedom for Non-Linear Molecules — 9701/12/M/J/222/Q37**[1]**

How many fundamental vibrational normal modes does a non-linear polyatomic molecule with N atoms possess?

- [A] $3N - 5$
- [B] $3N$
- [C] $3N - 6$
- [D] $2N + 1$

38 [HARD] Vibrational Modes of Water (H_2O) — 9701/13/M/J/223/Q38**[1]**

Water (H_2O) has $N = 3$ atoms and is non-linear. How many fundamental vibrational normal modes does it have, and what are they?

- [A] 3 vibrational modes: symmetric stretch, asymmetric stretch, and bending (scissoring).
- [B] 4 vibrational modes.
- [C] 6 vibrational modes.
- [D] 2 vibrational modes.

39 [EASY] Vibrational Modes of Carbon Dioxide (CO_2) — 9701/11/M/J/224/Q39**[1]**

Carbon dioxide (CO_2) is a linear triatomic molecule ($N = 3$). How many fundamental vibrational modes does it possess, and how many are IR-active?

- [A] 4 vibrational modes; all 4 are IR-active.
- [B] 3 vibrational modes; all 3 are IR-inactive.
- [C] 6 vibrational modes; 2 are IR-active.
- [D] 4 vibrational modes ($3N - 5 = 4$); 3 are IR-active (asymmetric stretch and two degenerate bending modes) and 1 is IR-inactive (symmetric stretch).

40 [HARD] Degenerate Vibrations Concept — 9701/12/M/J/220/Q40**[1]**

What does it mean for vibrational normal modes in a molecule (such as the two bending modes of CO_2) to be 'degenerate'?

- [A] They destroy the molecule.
- [B] They have identical vibrational frequencies and quantum energy levels because they occur along mutually perpendicular, structurally equivalent planes.
- [C] They produce zero radiation.
- [D] They occur only at absolute zero.

41 [EASY] Distinguishing Primary Amide from Carboxylic Acid — 9701/13/M/J/221/Q41**[1]**

Both ethanamide and ethanoic acid have carbonyl groups. How does the infrared spectrum of ethanamide cleanly distinguish it from ethanoic acid?

- [A] Ethanamide shows a single sharp peak at 2250 cm^{-1} .
- [B] Ethanamide shows two distinct N-H stretching spikes at $3180\text{--}3350 \text{ cm}^{-1}$ and an Amide I carbonyl peak at $1650\text{--}1680 \text{ cm}^{-1}$, whereas ethanoic acid shows an extremely broad O-H band across $2500\text{--}3300 \text{ cm}^{-1}$ and a C=O peak at 1715 cm^{-1} .
- [C] Ethanoic acid shows no peaks above 2000 cm^{-1} .
- [D] Both have identical spectra.

42 [HARD] Infrared Spectrum of Deuterated Compounds — 9701/11/M/J/222/Q42**[1]**

How does replacing hydrogen with deuterium in ethanol ($\text{CH}_3\text{CH}_2\text{OD}$ vs $\text{CH}_3\text{CH}_2\text{OH}$) affect the O-H stretching absorption band?

- [A] The O-D stretch shifts to 4000 cm^{-1} .
- [B] The peak disappears with zero absorption.
- [C] The peak shifts to the fingerprint region at 500 cm^{-1} .
- [D] The O-D stretch shifts to a significantly lower wavenumber ($\sim 2450\text{ cm}^{-1}$) compared to the O-H stretch ($\sim 3350\text{ cm}^{-1}$) due to the doubled mass of deuterium.

43 [EASY] Atmospheric Window (8 to 14 micrometers) — 9701/12/M/J/223/Q43**[1]**

What is the 'atmospheric infrared window' (wavenumbers 700 to 1250 cm^{-1} / wavelengths 8 to 14 micrometers)?

- [A] A spectral region where clean air (CO_2 and H_2O) absorbs very little thermal infrared radiation, allowing Earth's surface heat to escape directly into space.
- [B] A hole in the ozone layer over Antarctica.
- [C] A glass dome placed over weather stations.
- [D] A region of the atmosphere that emits visible light.

44 [HARD] Global Warming Potential of SF_6 and CFCs — 9701/13/M/J/224/Q44**[1]**

Why do chlorofluorocarbons (CFCs) and sulfur hexafluoride (SF_6) have Global Warming Potentials (GWPs) thousands of times higher than CO_2 ?

- [A] They absorb sunlight directly.
- [B] They produce carbon dioxide when exposed to moisture.
- [C] They possess intense C-F and S-F vibrational absorptions directly within the 8-14 micrometer atmospheric window and have atmospheric lifetimes of hundreds to thousands of years.
- [D] They react with nitrogen gas to form ozone.

45 [EASY] Infrared Spectrum of Methanol Vapor vs Liquid — 9701/11/M/J/220/Q45**[1]**

Why does gaseous methanol show a sharp, narrow O-H absorption band at 3680 cm^{-1} , whereas liquid methanol shows a broad band at 3340 cm^{-1} ?

- [A] In the gas phase, methanol molecules are isolated with zero intermolecular hydrogen bonding, yielding a single sharp vibrational frequency, whereas the liquid contains a chaotic network of hydrogen bonds.
- [B] Gaseous methanol has no chemical bonds.
- [C] Liquid methanol contains metallic bonds.
- [D] Gaseous methanol decomposes into carbon.

46 [HARD] Infrared Spectrum of Cycloalkanes: Ring Strain Effect — 9701/12/M/J/221/Q46**[1]**

Why does cyclopropanone exhibit its carbonyl stretch at an unusually high wavenumber (1815 cm^{-1}) compared to cyclohexanone (1715 cm^{-1})?

- [A] Cyclopropanone is an aromatic compound.
- [B] The carbonyl bond becomes a single bond.
- [C] Cyclopropanone forms hydrogen bonds.
- [D] Angle strain in the three-membered ring forces the exocyclic $\text{C}=\text{O}$ carbon to use orbitals with greater s-character ($-\text{sp}$), strengthening the $\text{C}=\text{O}$ bond and increasing its force constant.

47 [EASY] Identification of Secondary Alcohol Oxidation by IR — 9701/13/M/J/222/Q47**[1]**

A secondary alcohol is heated with acidified potassium dichromate(VI). How can IR spectroscopy confirm that oxidation has taken place?

- [A] A nitrile band appears at 2250 cm^{-1} .
- [B] The broad O-H band at 3300 cm^{-1} disappears, and a strong, sharp ketone carbonyl peak appears at 1715 cm^{-1} .
- [C] An ester C-O peak appears at 1250 cm^{-1} .
- [D] An alkene peak appears at 1650 cm^{-1} .

48 [HARD] IR Spectrum of Poly(ethene) vs Poly(chloroethene) — 9701/11/M/J/223/Q48**[1]**

How does the infrared spectrum of poly(chloroethene) (PVC) differ from that of poly(ethene)?

- [A] PVC shows a broad O-H band at 3300 cm^{-1} .
- [B] PVC shows a sharp carbonyl peak at 1735 cm^{-1} .
- [C] PVC shows intense C-Cl stretching bands in the $600\text{--}750\text{ cm}^{-1}$ region and modified C-H bending bands, whereas poly(ethene) shows only alkane C-H stretches and bends.
- [D] Poly(ethene) shows an alkene peak at 1650 cm^{-1} .

49 [EASY] Infrared Transparency of Salt Optics (NaCl / KBr) — 9701/12/M/J/224/Q49**[1]**

Why are sample cells and prisms in infrared spectrometers made of ionic salts like sodium chloride (NaCl) or potassium bromide (KBr) rather than standard optical glass?

- [A] Glass is an explosive material.
- [B] Glass is a covalent silicate network whose Si-O bonds strongly absorb infrared radiation across the entire mid-IR spectrum, whereas ionic salts with heavy ions have no covalent vibrations above 650 cm^{-1} and are completely transparent to IR.
- [C] Glass reacts with all organic liquids.
- [D] NaCl prisms are completely flexible.

50 [HARD] Deducing Functional Group: Peak at 1690 cm^{-1} — 9701/13/M/J/220/Q50**[1]**

An organic compound displays a prominent infrared absorption at 1690 cm^{-1} . Which of the following functional groups is consistent with this absorption?

- [A] An α,β -unsaturated ketone or a secondary amide
- [B] An aliphatic ester
- [C] An acyl chloride
- [D] A saturated alkane

51 [EASY] IR Spectrum of Carboxylic Acid Dimer — 9701/11/M/J/221/Q51**[1]**

Why does ethanoic acid show an exceptionally broad, ragged O-H stretching band spanning $2500\text{ to }3300\text{ cm}^{-1}$ in pure liquid and concentrated solution?

- [A] Ethanoic acid is an ionic mineral acid.
- [B] Ethanoic acid decomposes into water and carbon.
- [C] Ethanoic acid contains double bonds between oxygen atoms.
- [D] Ethanoic acid forms stable, cyclic eight-membered hydrogen-bonded dimers with two strong hydrogen bonds that severely weaken and broaden the O-H stretching vibrations.

52 [HARD] Infrared Transmittance vs Absorbance — 9701/12/M/J/222/Q52**[1]**

What is the mathematical relationship between percentage transmittance (%T) and absorbance (A) in an infrared spectrum?

- [A] $A = 100 - \%T$
- [B] $A = \%T / 100$
- [C] $A = 2 - \log_{10}(\%T)$ (or $A = -\log_{10}(T)$)
- [D] $A = \log_{10}(\%T)$

53 [EASY] Baseline Drift in IR Spectra: Scattering — 9701/13/M/J/223/Q53**[1]**

What causes a sloping, drifting baseline in the infrared spectrum of a solid sample prepared as a KBr pellet?

- [A] Light scattering (Rayleigh and Mie scattering) caused by solid sample particles that were not ground finely enough (particle size comparable to IR wavelength, $\sim 2\text{--}15$ micrometers).
- [B] The presence of water on the detector.
- [C] The sample is undergoing nuclear decay.
- [D] The sample is an ester.

54 [HARD] IR Spectrum of Propanoic Acid vs Methyl Ethanoate — 9701/11/M/J/224/Q54**[1]**

Both propanoic acid and methyl ethanoate have the molecular formula $C_3H_6O_2$. How can they be distinguished unambiguously by infrared spectroscopy?

- [A] Methyl ethanoate shows a broad band at 3400 cm^{-1} .
- [B] Propanoic acid shows a sharp peak at 2250 cm^{-1} .
- [C] Propanoic acid shows a very broad, ragged O-H band across $2500\text{--}3300\text{ cm}^{-1}$ and a C=O peak at 1715 cm^{-1} , whereas methyl ethanoate shows NO O-H band and an ester C=O peak at 1740 cm^{-1} .
- [D] Both spectra are completely identical.

55 [EASY] Infrared Spectrum of Benzaldehyde — 9701/12/M/J/220/Q55**[1]**

What combination of infrared absorption bands confirms the structure of benzaldehyde (C_6H_5CHO)?

- [A] A broad O-H band at 3300 cm^{-1} .
- [B] A sharp aromatic C=C band at 1595 cm^{-1} , an aldehyde C=O stretch at 1700 cm^{-1} , and the aldehyde C-H Fermi doublet at 2730 cm^{-1} and 2820 cm^{-1} .
- [C] A sharp nitrile peak at 2250 cm^{-1} .
- [D] An ester C-O peak at 1200 cm^{-1} .

SUBTOPIC 22.2 — MASS SPECTROMETRY: MOLECULAR IONS, [M+1], [M+2] & FRAGMENTATION (Q56 – Q100)**56 [HARD]** Carbon Counting using the [M+1] Peak — 9701/11/M/J/23/Q39**[1]**

The mass spectrum of an unknown hydrocarbon shows a molecular ion peak, M, at $m/z = 72$ with a relative abundance of 80.0%. It also shows an [M+1] peak at $m/z = 73$ with a relative abundance of 4.4%.

How many carbon atoms are present in one molecule of this hydrocarbon?

- [A] 4 carbon atoms
- [B] 6 carbon atoms
- [C] 7 carbon atoms
- [D] 5 carbon atoms

57 [EASY] Chlorine Isotope Ratio: [M] and [M+2] Peaks — 9701/12/M/J/23/Q39**[1]**

The mass spectrum of a mono-halogenated alkane displays a molecular ion peak at $m/z = 78$ and an [M+2] peak at $m/z = 80$ in a relative abundance ratio of approximately 3 : 1.

Which halogen is present in this molecule?

- [A] One bromine atom (due to natural isotopes ^{79}Br and ^{81}Br in a 1:1 ratio)
- [B] One iodine atom (which is monoisotopic, ^{127}I)
- [C] One chlorine atom (due to natural isotopes ^{35}Cl and ^{37}Cl in approximately a 3:1 ratio)
- [D] One fluorine atom (which is monoisotopic, ^{19}F)

58 [EASY] Bromine Isotope Ratio: [M] and [M+2] Peaks — 9701/13/M/J/23/Q39**[1]**

The mass spectrum of an organic compound shows two molecular ion peaks of virtually EQUAL height at $m/z = 108$ and $m/z = 110$.

What is the identity of this compound?

- [A] Bromoethane, $\text{CH}_3\text{CH}_2\text{Br}$
- [B] Chloroethane, $\text{CH}_3\text{CH}_2\text{Cl}$
- [C] Iodoethane, $\text{CH}_3\text{CH}_2\text{I}$
- [D] 1,2-dichloroethane, $\text{CH}_2\text{ClCH}_2\text{Cl}$

59 [MEDIUM] Two Chlorine Atoms: $[M] : [M+2] : [M+4]$ Peak Ratio — 9701/11/O/N/23/Q39**[1]**

A compound contains TWO chlorine atoms per molecule (e.g. dichloromethane, CH_2Cl_2).

What is the theoretical relative peak height ratio of its molecular ion cluster $[M] : [M+2] : [M+4]$?

- [A] 3 : 2 : 1
- [B] 1 : 2 : 1
- [C] 1 : 1 : 1
- [D] 9 : 6 : 1

60 [HARD] Two Bromine Atoms Peak Ratio — 9701/11/M/J/220/Q60**[1]**

A compound contains TWO bromine atoms per molecule (e.g. 1,2-dibromoethane). What is the relative peak height ratio of its molecular ion cluster $[M] : [M+2] : [M+4]$?

- [A] 9 : 6 : 1
- [B] 1 : 2 : 1
- [C] 1 : 1 : 1
- [D] 3 : 1 : 1

61 [EASY] One Chlorine and One Bromine Atom Peak Ratio — 9701/12/M/J/221/Q61**[1]**

A compound contains ONE chlorine atom and ONE bromine atom per molecule (e.g. 1-bromo-2-chloroethane). What is the relative peak height ratio of its $[M] : [M+2] : [M+4]$ cluster?

- [A] 9 : 6 : 1
- [B] 3 : 4 : 1
- [C] 1 : 2 : 1
- [D] 3 : 1 : 1

62 [HARD] Characteristic Fragmentation Peak: $m/z = 15$ — 9701/13/M/J/222/Q62**[1]**

In the mass spectrum of propane, what chemical species is responsible for the fragmentation peak observed at $m/z = 15$?

- [A] A neutral methyl radical, $\text{CH}_3^{\bullet}$
- [B] A methane molecule, CH_4
- [C] A carbon atom, C^+
- [D] The methyl carbocation, $[\text{CH}_3]^+$

63 [EASY] Characteristic Fragmentation Peak: $m/z = 29$ — 9701/11/M/J/223/Q63**[1]**

In the mass spectrum of propanal ($\text{CH}_3\text{CH}_2\text{CHO}$, $M_r = 58$), what two different fragment ions can both contribute to the prominent peak at $m/z = 29$?

- [A] The ethyl cation, $[\text{CH}_3\text{CH}_2]^+$, and the formyl (acylium) cation, $[\text{CHO}]^+$
- [B] The methyl cation and hydroxyl cation
- [C] The propanal molecular ion
- [D] Two carbon atoms

64 [HARD] Characteristic Fragmentation Peak: $m/z = 43$ — 9701/12/M/J/224/Q64**[1]**

The mass spectrum of an unknown carbonyl compound shows a base peak (tallest peak) at $m/z = 43$. Which of the following ions is responsible for this peak in a methyl ketone?

- [A] The propyl cation exclusively
- [B] The methoxy cation
- [C] The ethanoyl (acylium) cation, $[\text{CH}_3\text{CO}]^+$
- [D] The carboxyl cation

65 [EASY] Characteristic Fragmentation Peak: $m/z = 57$ — 9701/13/M/J/220/Q65

[1]

In the mass spectrum of 2-methylpropane (isobutane), what fragment ion produces the dominant base peak at $m/z = 57$?

- [A] The tert-butyl carbocation, $[(CH_3)_3C]^+$, formed by the loss of a hydrogen atom ($M - 1$)
- [B] The ethyl cation
- [C] The methyl cation
- [D] The propane molecular ion

66 [HARD] Characteristic Fragmentation Peak: $m/z = 77$ — 9701/11/M/J/221/Q66

[1]

In the mass spectrum of an aromatic organic compound (such as ethylbenzene), what ion produces the characteristic fragmentation peak at $m/z = 77$?

- [A] The tropylium ion
- [B] The benzyl radical
- [C] A benzene molecule
- [D] The phenyl carbocation, $[C_6H_5]^+$

67 [EASY] Characteristic Fragmentation Peak: $m/z = 91$ (Tropylium Ion) — 9701/12/M/J/222/Q67

[1]

Why is the base peak in the mass spectrum of alkylbenzenes containing a benzyl group (such as ethylbenzene or propylbenzene) almost always at $m/z = 91$?

- [A] m/z 91 is a neutral radical.
- [B] Benzylic cleavage forms the benzyl cation $[C_6H_5CH_2]^+$ ($m/z = 91$), which rapidly rearranges into the exceptionally stable, aromatic seven-membered cycloheptatrienyl (tropylium) cation $[C_7H_7]^+$.
- [C] The benzene ring explodes into fragments.
- [D] m/z 91 represents a toluene molecule.

68 [HARD] Primary Alcohol Fragmentation: $m/z = 31$ — 9701/13/M/J/223/Q68

[1]

The mass spectrum of an unknown aliphatic alcohol shows a base peak at $m/z = 31$. What ion is this, and what does it indicate?

- [A] The methyl cation, $[CH_3]^+$
- [B] The ethyl cation, $[C_2H_5]^+$
- [C] The hydroxymethyl cation, $[CH_2OH]^+$, which is characteristic of primary alcohols ($R-CH_2OH$).
- [D] The hydroxyl ion, $[OH]^-$

69 [EASY] Secondary Alcohol Fragmentation: Loss of Alkyl Groups — 9701/11/M/J/224/Q69

[1]

Propan-2-ol ($CH_3CH(OH)CH_3$, $M_r = 60$) shows a prominent fragment peak at $m/z = 45$. What fragmentation process produces this ion?

- [A] Loss of water ($M - 18$)
- [B] Alpha-cleavage with loss of a methyl radical ($\cdot CH_3$, mass 15), leaving the resonance-stabilized secondary oxonium ion $[CH_3CH=OH]^+$ ($m/z = 60 - 15 = 45$).
- [C] Loss of hydrogen gas
- [D] Cleavage of both methyl groups simultaneously

70 [HARD] Alcohol Dehydration Fragmentation: Peak at $[M - 18]$ — 9701/12/M/J/220/Q70

[1]

The mass spectrum of an alcohol exhibits a prominent peak at $m/z = [M - 18]$. What neutral small molecule was eliminated from the molecular ion?

- [A] A neutral water molecule (H_2O , mass 18)
- [B] An ammonium ion
- [C] A fluorine atom
- [D] An ozone molecule

71 [EASY] Calculating Number of Carbons from $[M]$ and $[M+1]$ (Hexane) — 9701/13/M/J/221/Q71**[1]**

A hydrocarbon displays a molecular ion peak at $m/z = 86$ with relative intensity 65.0%, and an $[M+1]$ peak at $m/z = 87$ with relative intensity 4.3%.

What is the molecular formula of the compound?

- [A] C_5H_{12}
- [B] C_7H_{16}
- [C] C_6H_{12}
- [D] C_6H_{14}

72 [HARD] Calculating Number of Carbons from $[M]$ and $[M+1]$ (Benzene) — 9701/11/M/J/222/Q72**[1]**

A compound has a molecular ion peak at $m/z = 78$ with abundance 100%, and an $[M+1]$ peak at $m/z = 79$ with abundance 6.6%.
How many carbon atoms are in the molecule?

- [A] 5 carbon atoms
- [B] 7 carbon atoms
- [C] 6 carbon atoms
- [D] 8 carbon atoms

73 [EASY] Identification of Carboxylic Acid: $m/z = 45$ — 9701/12/M/J/223/Q73**[1]**

In the mass spectrum of propanoic acid (CH_3CH_2COOH , $M_r = 74$), what fragment ion produces the peak at $m/z = 45$?

- [A] The carboxyl cation, $[COOH]^+$
- [B] The ethyl cation
- [C] The propanoyl ion
- [D] The methyl cation

74 [HARD] Carboxylic Acid Fragmentation: Loss of $-COOH$ ($m/z = [M - 45]$) — 9701/13/M/J/224/Q74**[1]**

In the mass spectrum of butanoic acid ($M_r = 88$), what fragment ion is responsible for the peak at $m/z = 43$ (which corresponds to $[M - 45]$)?

- [A] The carboxyl cation
- [B] The methyl cation
- [C] The propyl carbocation, $[C_3H_7]^+$ (mass $88 - 45 = 43$)
- [D] The butyl cation

75 [EASY] McLafferty Rearrangement in Carbonyls — 9701/11/M/J/220/Q75**[1]**

What is the McLafferty rearrangement in the mass spectrometry of carbonyl compounds containing a gamma-hydrogen atom (e.g. pentan-2-one)?

- [A] A spontaneous explosion of the molecule.
- [B] An intramolecular transfer of a gamma-hydrogen atom to the carbonyl oxygen via a six-membered cyclic transition state, followed by beta-cleavage to eliminate a neutral alkene molecule (e.g. ethene).
- [C] The loss of carbon dioxide.
- [D] The conversion of the ketone into an alkane.

76 [HARD] Base Peak vs Molecular Ion Peak — 9701/12/M/J/221/Q76**[1]**

What is the fundamental difference between the 'molecular ion peak' (M) and the 'base peak' in a mass spectrum?

- [A] The base peak is always the heaviest peak.
- [B] The molecular ion peak is always the tallest peak.
- [C] The base peak represents the solvent.
- [D] The molecular ion peak corresponds to the unfragmented radical cation $[M]^+ \cdot$ and gives the relative molecular mass (M_r), whereas the base peak is the most intense (tallest) peak assigned 100% relative abundance.

77 [EASY] High-Resolution Mass Spectrometry (HRMS) — 9701/13/M/J/222/Q77**[1]**

How does High-Resolution Mass Spectrometry (HRMS) distinguish between compounds with identical nominal integer molecular masses, such as propanone ($\text{C}_3\text{H}_6\text{O}$, $M_r = 58$) and butane (C_4H_{10} , $M_r = 58$)?

- [A] HRMS measures the color of the molecule.
- [B] HRMS uses a thermometer.
- [C] HRMS measures m/z values to four or five decimal places, detecting tiny differences in exact isotopic mass ($\text{C}_3\text{H}_6\text{O} = 58.0419$ vs $\text{C}_4\text{H}_{10} = 58.0782$).
- [D] HRMS measures the boiling point to five decimals.

78 [HARD] Ionization Methods: Electron Impact (EI) — 9701/11/M/J/223/Q78**[1]**

In a standard electron impact (EI) mass spectrometer, what happens when high-energy electrons (typically 70 eV) collide with gaseous sample molecules?

- [A] An electron is knocked out of the sample molecule to form a positively charged radical cation: $\text{M}(\text{g}) + \text{e}^- \rightarrow [\text{M}]^{+\bullet}(\text{g}) + 2\text{e}^-$.
- [B] The molecule absorbs an electron to form an anion.
- [C] The molecule is converted into gold.
- [D] The molecule is completely burned to CO_2 .

79 [EASY] Role of the Magnetic Sector Analyzer — 9701/12/M/J/224/Q79**[1]**

How does a magnetic sector in a mass spectrometer separate ions of different mass-to-charge ratios (m/z)?

- [A] Lighter ions are stopped by a filter.
- [B] Heavier ions are burned by the magnet.
- [C] All ions travel in straight lines.
- [D] A perpendicular magnetic field deflects moving positive ions into circular paths whose radius of curvature is proportional to the square root of m/z ; lighter ions are deflected more than heavier ions.

80 [HARD] Why Neutral Fragments are Not Detected — 9701/13/M/J/220/Q80**[1]**

Why are neutral molecules and neutral free radicals formed during fragmentation (such as $\cdot\text{CH}_3$ or H_2O) NOT recorded on a mass spectrum?

- [A] They are absorbed by the glass walls.
- [B] They carry zero electrical charge and are therefore neither accelerated by electric fields nor deflected by magnetic fields.
- [C] They are invisible to light.
- [D] They travel at the speed of light.

81 [EASY] Nitrogen Rule in Mass Spectrometry — 9701/11/M/J/221/Q81**[1]**

According to the 'Nitrogen Rule', what does an ODD nominal molecular ion mass (e.g. $m/z = 45, 59, 73$) indicate about an organic compound containing only C, H, N, and O?

- [A] The compound contains an even number of nitrogen atoms.
- [B] The compound contains an ODD number of nitrogen atoms (e.g. 1, 3, 5 nitrogens).
- [C] The compound contains no nitrogen atoms.
- [D] The compound is an alkane.

82 [HARD] Mass Spectrum of Branched vs Linear Alkanes — 9701/12/M/J/222/Q82**[1]**

Why does the molecular ion peak $[\text{M}]^{+\bullet}$ of a branched alkane (such as 2,2-dimethylpropane / neopentane) have an extremely low or undetectable intensity compared to linear pentane?

- [A] Branched alkanes cannot be vaporized.
- [B] Branched alkanes do not lose electrons.
- [C] Neopentane has a higher molecular mass.
- [D] Branching creates highly substituted quaternary carbons where C-C bond cleavage is extraordinarily fast, producing stable tertiary or tert-butyl carbocations that fragment the molecular ion instantly.

83 [EASY] Mass Spectrum of Esters: Cleavage Across Ester Linkage — 9701/13/M/J/223/Q83**[1]**

In the mass spectrum of ethyl ethanoate ($\text{CH}_3\text{COOCH}_2\text{CH}_3$, $M_r = 88$), what fragment ion produces the peak at $m/z = 43$, and what produces the peak at $m/z = 29$?

- [A] $m/z = 43$ is the acylium ion $[\text{CH}_3\text{CO}]^+$; $m/z = 29$ is the ethyl cation $[\text{CH}_3\text{CH}_2]^+$.
- [B] $m/z = 43$ is propane; $m/z = 29$ is methanoic acid.
- [C] $m/z = 43$ is $[\text{C}_3\text{H}_7]^+$; $m/z = 29$ is $[\text{CHO}]^+$.
- [D] Both peaks are due to carbon atoms.

84 [HARD] Three Chlorine Atoms: Peak Ratio for $[M]$ to $[M+6]$ — 9701/11/M/J/224/Q84**[1]**

What is the relative abundance ratio of the molecular ion cluster $[M] : [M+2] : [M+4] : [M+6]$ for a compound containing THREE chlorine atoms (e.g. trichloromethane, CHCl_3)?

- [A] 9 : 6 : 1 : 1
- [B] 3 : 3 : 1 : 1
- [C] 27 : 27 : 9 : 1
- [D] 1 : 3 : 3 : 1

85 [EASY] Three Bromine Atoms: Peak Ratio for $[M]$ to $[M+6]$ — 9701/12/M/J/220/Q85**[1]**

What is the relative abundance ratio of the molecular ion cluster $[M] : [M+2] : [M+4] : [M+6]$ for a compound containing THREE bromine atoms (e.g. tribromomethane, CHBr_3)?

- [A] 1 : 3 : 3 : 1
- [B] 27 : 27 : 9 : 1
- [C] 1 : 2 : 2 : 1
- [D] 3 : 3 : 1 : 1

86 [HARD] Deducing Halogen Count from Mass Spectrum — 9701/13/M/J/221/Q86**[1]**

The molecular ion cluster of an unknown haloalkane shows four peaks at $m/z = 170, 172, 174$, and 176 in the approximate ratio of 27 : 27 : 9 : 1. What halogens are present in this molecule?

- [A] Three bromine atoms
- [B] Two chlorines and one bromine
- [C] One chlorine and two bromines
- [D] Three chlorine atoms

87 [EASY] Loss of Methyl Radical: Peak at $[M - 15]$ — 9701/11/M/J/222/Q87**[1]**

The mass spectrum of an unknown ketone shows a molecular ion peak at $m/z = 86$ and a prominent fragment peak at $m/z = 71$. What fragmentation step occurred to form the $m/z = 71$ peak?

- [A] Loss of water (mass 18)
- [B] Loss of a neutral methyl radical ($\cdot\text{CH}_3$, mass 15) via alpha-cleavage: $86 - 15 = 71$.
- [C] Loss of carbon monoxide (mass 28)
- [D] Loss of hydrogen gas

88 [HARD] Loss of Ethyl Radical: Peak at $[M - 29]$ — 9701/12/M/J/223/Q88**[1]**

In the mass spectrum of 3-pentanone ($\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$, $M_r = 86$), what ion produces the base peak at $m/z = 57$?

- [A] The butyl cation
- [B] The methyl cation
- [C] The propionyl (propanoyl) cation, $[\text{CH}_3\text{CH}_2\text{CO}]^+$, formed by the loss of an ethyl radical ($\cdot\text{CH}_2\text{CH}_3$, mass 29): $86 - 29 = 57$.
- [D] The acetone cation

89 [EASY] Identification of Alkene by Fragmentation — 9701/13/M/J/224/Q89

[1]

The mass spectrum of an unknown hydrocarbon shows a molecular ion peak at $m/z = 56$ and a base peak at $m/z = 41$. What is the formula of the hydrocarbon and the identity of the $m/z = 41$ fragment ion?

- [A] C_3H_8 (propane); $[CH_3]^+$
- [B] C_4H_8 (butene); the fragment ion is the resonance-stabilized allyl cation, $[CH_2=CH-CH_2]^+$ ($m/z = 41$).
- [C] C_4H_{10} (butane); $[C_2H_5]^+$
- [D] C_5H_{12} (pentane); $[C_3H_7]^+$

90 [HARD] Sulfur-34 Isotope in Mass Spectrometry — 9701/11/M/J/220/Q90

[1]

Naturally occurring sulfur has a minor isotope, ^{34}S , with an abundance of ~4.2%.

What peak pattern in the molecular ion cluster identifies a compound containing ONE sulfur atom (such as ethanethiol, C_2H_5SH)?

- [A] A distinct $[M+2]$ peak with an abundance of approximately 4.4% relative to the molecular ion peak $[M]$.
- [B] An $[M+2]$ peak of 33% intensity.
- [C] An $[M+2]$ peak of 100% intensity.
- [D] An $[M+1]$ peak of 50% intensity.

91 [EASY] Distinguishing Isomers by Fragmentation: Propan-1-ol vs Propan-2-ol — 9701/12/M/J/221/Q91

[1]

How does the mass spectrum of propan-1-ol ($CH_3CH_2CH_2OH$) differ from that of propan-2-ol ($CH_3CH(OH)CH_3$)?

- [A] Propan-1-ol has a molecular ion at $m/z = 60$; propan-2-ol has $M_r = 70$.
- [B] Propan-2-ol shows no fragmentation.
- [C] Propan-1-ol shows a peak at $m/z = 100$.
- [D] Propan-1-ol has its base peak at $m/z = 31$ ($[CH_2OH]^+$), whereas propan-2-ol has its base peak at $m/z = 45$ ($[CH_3CHOH]^+$).

92 [HARD] Distinguishing Isomers: Pentan-2-one vs Pentan-3-one — 9701/13/M/J/222/Q92

[1]

How can mass spectrometry distinguish pentan-2-one from pentan-3-one (both have $M_r = 86$)?

- [A] Pentan-2-one has a higher molecular ion mass.
- [B] Pentan-3-one does not fragment.
- [C] Pentan-2-one displays prominent fragment peaks at $m/z = 43$ ($[CH_3CO]^+$) and $m/z = 58$ (McLafferty rearrangement), whereas pentan-3-one displays a base peak at $m/z = 57$ ($[CH_3CH_2CO]^+$).
- [D] Pentan-2-one shows peaks only above $m/z = 100$.

93 [EASY] Determination of Empirical and Molecular Formula from Data — 9701/11/M/J/223/Q93

[1]

Elemental combustion analysis of an organic compound gives 54.5% C, 9.1% H, and 36.4% O by mass. Its mass spectrum shows a molecular ion peak at $m/z = 88$.

What is its molecular formula?

- [A] $C_4H_8O_2$
- [B] C_2H_4O
- [C] $C_3H_6O_3$
- [D] $C_5H_{12}O$

94 [HARD] Mass Spectrometry: Detection of Radicals vs Cations — 9701/12/M/J/224/Q94

[1]

In a mass spectrometer, a molecular ion $[CH_3CH_2Cl]^+$ cleaves into $[CH_3CH_2]^+$ and *Cl . Which of these two particles arrives at the detector and produces a peak?

- [A] Both particles produce peaks.
- [B] Only the neutral chlorine radical.
- [C] Only the ethyl carbocation, $[CH_3CH_2]^+$ ($m/z = 29$); the neutral chlorine radical (*Cl) is unaffected by the magnetic field and is pumped away.
- [D] Neither particle reaches the detector.

95 [EASY] Electrospray Ionization (ESI) Concept — 9701/13/M/J/220/Q95**[1]**

Why is Electrospray Ionization (ESI) classified as a 'soft' ionization technique compared to Electron Impact (EI)?

- [A] ESI burns the molecule completely.
- [B] ESI produces intact protonated molecular ions ($[M+H]^+$) with virtually zero fragmentation, allowing accurate determination of the molecular mass of delicate, large biomolecules like proteins.
- [C] ESI uses high-voltage lasers to break all bonds.
- [D] ESI detects only neutral atoms.

96 [HARD] Calculation of Number of Carbons: $[M+1]$ Formula Derivation — 9701/11/M/J/221/Q96**[1]**

Why does the formula for calculating carbon atoms, $n = (100 / 1.1) \times (\text{abundance } [M+1] / \text{abundance } M)$, divide by 1.1?

- [A] Because 1.1 is the mass of an electron.
- [B] Because 1.1 is the speed of light in vacuum.
- [C] Because hydrogen has an abundance of 1.1%.
- [D] Because each individual carbon atom has an independent natural probability of approximately 1.1% of being the carbon-13 (^{13}C) isotope.

97 [EASY] Metastable Ions in Mass Spectrometry — 9701/12/M/J/222/Q97**[1]**

What is a 'metastable ion' in mass spectrometry, and how does it appear in the spectrum?

- [A] An ion that lasts forever.
- [B] An uncharged neutral particle.
- [C] An ion that fragments during flight through the field-free region between the acceleration plates and magnetic sector, producing a broad, diffuse peak at a non-integer apparent mass $m^* = (m_2)^2 / m_1$.
- [D] An ion that emits green light.

98 [HARD] Identification of Unknown from Combined IR and Mass Spec — 9701/13/M/J/223/Q98**[1]**

An unknown liquid displays an IR absorption at 1715 cm^{-1} (no absorption above 3000 cm^{-1}). Its mass spectrum shows a molecular ion peak at $m/z = 58$ and a base peak at $m/z = 43$.

What is the compound?

- [A] Propanone, CH_3COCH_3
- [B] Propanal, $\text{CH}_3\text{CH}_2\text{CHO}$
- [C] Butane, C_4H_{10}
- [D] Ethanoic acid, CH_3COOH

99 [EASY] Identification of Haloalkane from Combined Data — 9701/11/M/J/224/Q99**[1]**

An organic compound displays no significant IR absorption above 3000 cm^{-1} . Its mass spectrum shows molecular ion peaks of equal height at $m/z = 122$ and $m/z = 124$, and a base peak at $m/z = 43$.

What is the compound?

- [A] 1-chlorobutane
- [B] Bromoethane
- [C] Iodopropane
- [D] 1-bromopropane (or 2-bromopropane), $\text{C}_3\text{H}_7\text{Br}$

100 [HARD] Tandem Mass Spectrometry (MS/MS) — 9701/12/M/J/220/Q100**[1]**

What is the primary function of Tandem Mass Spectrometry (MS/MS)?

- [A] It measures two samples at the same time.
- [B] The first mass analyzer selects a specific precursor ion of interest, which is fragmented by collision-induced dissociation (CID), and the second analyzer analyzes the resulting product fragments for unambiguous structural sequencing.
- [C] It uses two different magnetic fields to weigh the instrument.
- [D] It doubles the speed of the vacuum pump.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)

101 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 1: Carbon Atom Count from $[M+1]$ Peak Height — 9701/11/M/J/22/Q36**[1]**

In the mass spectrum of an organic compound, the molecular ion peak M has a relative abundance of 50.0%, and the $[M+1]$ peak has a relative abundance of 2.2%.

How many carbon atoms are present in one molecule of this compound?

- [A] 5 carbon atoms
- [B] 3 carbon atoms
- [C] 4 carbon atoms
- [D] 6 carbon atoms

102 [EASY] [HIGH-FREQUENCY EXAM REPEAT] HF 2: Identifying Chlorine vs Bromine from $[M+2]$ Ratio — 9701/12/M/J/22/Q36**[1]**

Which row correctly pairs the halogen with the observed relative abundance ratio of its $[M] : [M+2]$ molecular ion peaks?

- [A] One Chlorine atom: $[M] : [M+2] = 3 : 1$; One Bromine atom: $[M] : [M+2] = 1 : 1$
- [B] One Chlorine atom: $[M] : [M+2] = 1 : 1$; One Bromine atom: $[M] : [M+2] = 3 : 1$
- [C] One Chlorine atom: $[M] : [M+2] = 9 : 6$; One Bromine atom: $[M] : [M+2] = 1 : 2$
- [D] One Chlorine atom: $[M] : [M+2] = 1 : 3$; One Bromine atom: $[M] : [M+2] = 1 : 1$

103 [MEDIUM] [HIGH-FREQUENCY EXAM REPEAT] HF 3: Distinguishing Alcohol from Carboxylic Acid in IR — 9701/13/M/J/22/Q36**[1]**

How can an examiner unambiguously distinguish an alcohol from a carboxylic acid using only infrared spectroscopy?

- [A] An alcohol shows a sharp peak at 1715 cm^{-1} ; a carboxylic acid shows no peaks above 1500 cm^{-1} .
- [B] An alcohol shows a doublet at $3300\text{--}3500\text{ cm}^{-1}$; a carboxylic acid shows a single peak.
- [C] Both have completely identical infrared spectra.
- [D] An alcohol shows a smooth, broad O-H band at $3200\text{--}3600\text{ cm}^{-1}$ and NO carbonyl peak, whereas a carboxylic acid shows a very broad, ragged O-H band spanning $2500\text{--}3300\text{ cm}^{-1}$ overlapping C-H stretches, alongside a sharp C=O peak at $1700\text{--}1725\text{ cm}^{-1}$.

104 [MEDIUM] [HIGH-FREQUENCY EXAM REPEAT] HF 4: Ester vs Ketone Carbonyl Frequency in IR — 9701/11/O/N/22/Q36**[1]**

Why does the carbonyl (C=O) absorption in an ester (e.g. ethyl ethanoate) appear at a HIGHER wavenumber ($\sim 1740\text{ cm}^{-1}$) than in an aliphatic ketone (e.g. propanone, $\sim 1715\text{ cm}^{-1}$)?

- [A] Esters contain hydrogen bonding.
- [B] The electronegative ester oxygen atom withdraws electron density from the carbonyl group inductively ($-I$ effect), strengthening the C=O bond and increasing its force constant.
- [C] Ketones have higher molecular masses.
- [D] The C=O bond in an ester is a single bond.

105 [EASY] [HIGH-FREQUENCY EXAM REPEAT] HF 5: Nitrile Group Identification in IR — 9701/12/O/N/22/Q36**[1]**

An unknown organic liquid gives a sharp, medium-intensity absorption band at 2250 cm^{-1} in its infrared spectrum. Which functional group is present?

- [A] A nitrile group, $\text{C}\equiv\text{N}$
- [B] An alcohol group, OH
- [C] A carbonyl group, $\text{C}=\text{O}$
- [D] An alkene group, $\text{C}=\text{C}$

106 [MEDIUM] [HIGH-FREQUENCY EXAM REPEAT] HF 6: Peak Ratio for Two Chlorine Atoms — 9701/13/O/N/22/Q36**[1]**

A dichlorinated hydrocarbon (e.g. CH_2Cl_2) exhibits three molecular ion peaks in its mass spectrum at $m/z = 84, 86$, and 88 . What is the theoretical ratio of heights for these three peaks $[M] : [M+2] : [M+4]$?

- [A] 3 : 2 : 1
- [B] 1 : 2 : 1
- [C] 9 : 6 : 1
- [D] 27 : 9 : 1

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] HF 7: Base Peak at $m/z = 43$ in Methyl Ketone — 9701/11/M/J/21/Q36**[1]**

The mass spectrum of propan-2-one (CH_3COCH_3 , $M_r = 58$) has its base peak at $m/z = 43$. What is the formula and structure of the ion responsible for this peak?

- [A] $[\text{C}_3\text{H}_7]^+$ (propyl cation)
- [B] $[\text{CH}_3\text{CO}]^+$ (ethanoyl / acetyl cation)
- [C] $[\text{CHO}]^+$ (formyl cation)
- [D] $[\text{C}_2\text{H}_3\text{O}_2]^+$ (acetate cation)

108 [EASY] [HIGH-FREQUENCY EXAM REPEAT] HF 8: Greenhouse Gases: Condition for IR Absorption — 9701/12/M/J/21/Q36**[1]**

What fundamental condition must be met for a vibrational mode of an atmospheric gas molecule to absorb infrared radiation?

- [A] The molecule must contain an odd number of atoms.
- [B] The molecule must be in the liquid state.
- [C] The molecule must be an element rather than a compound.
- [D] The vibration must produce a change in the molecular dipole moment of the molecule.

109 [MEDIUM] [HIGH-FREQUENCY EXAM REPEAT] HF 9: Primary vs Secondary Amine N-H Absorption in IR — 9701/13/M/J/21/Q36**[1]**

How does the N-H stretching absorption in the infrared spectrum of a primary amine (R-NH_2) compare with that of a secondary amine (R_2NH)?

- [A] A primary amine exhibits two distinct sharp peaks (a doublet) between 3300 and 3500 cm^{-1} , whereas a secondary amine exhibits only one sharp peak.
- [B] A primary amine exhibits no peaks above 3000 cm^{-1} ; a secondary amine exhibits three peaks.
- [C] A primary amine shows a broad band from 2500 to 3000 cm^{-1} ; a secondary amine shows a sharp peak at 1715 cm^{-1} .
- [D] Both exhibit identical single peaks.

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 10: Deduce Structure from Combined IR and Mass Spectrum — 9701/11/O/N/21/Q36**[1]**

An unknown organic compound has a molecular ion peak at $m/z = 60$ with $[M+1]$ at $m/z = 61$ (abundance ratio indicates 3 carbons). Its infrared spectrum shows a smooth, broad absorption band at 3350 cm^{-1} and no absorption between 1650 and 1750 cm^{-1} . Its mass spectrum shows a base peak at $m/z = 31$.

What is the systematic IUPAC name of the compound?

- [A] Propan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- [B] Ethanoic acid, CH_3COOH
- [C] Propan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- [D] Methoxymethane, CH_3OCH_3

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------

COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. An alcohol exhibits a hydrogen-bonded O-H stretch as a smooth, broad parabolic band at 3200–3600 cm ⁻¹ without a carbonyl peak. In contrast, a carboxylic acid exhibits an exceptionally broad, ragged O-H stretching band spanning 2500–3300 cm ⁻¹ (caused by very strong hydrogen bonding in carboxylic acid dimers) that overlaps the C-H stretching vibrations around 2950 cm ⁻¹ , accompanied by an intense carbonyl (C=O) stretch at 1700–1725 cm ⁻¹ .	[1]
2	D	Option D is correct. A molecule absorbs infrared radiation only if a vibrational mode (stretch or bend) causes a net change in its molecular dipole moment, allowing interaction with the oscillating electric field of IR light. In CO ₂ , asymmetric stretching and bending vibrations create dynamic dipoles that absorb IR strongly. In contrast, symmetrical homonuclear diatomic molecules like N ₂ and O ₂ have zero dipole moment at all times, making them IR-inactive.	[1]
3	A	Option A is correct. The presence of a sharp carbonyl stretch at 1735–1750 cm ⁻¹ and strong C–O single-bond stretches at 1150–1250 cm ⁻¹ , combined with the total absence of any O–H absorption (2500–3600 cm ⁻¹), conclusively identifies an ester (R–COO–R'). The formula C ₃ H ₆ O ₂ corresponds to methyl ethanoate (CH ₃ COOCH ₃) or ethyl methanoate (HCOOCH ₂ CH ₃).	[1]
4	C	Option C is correct. Above 1500 cm ⁻¹ , characteristic functional groups (C=O, O–H, N–H, C#N) exhibit diagnostic group frequencies. Below 1500 cm ⁻¹ (the fingerprint region), complex skeletal stretching and bending vibrations of the entire molecular framework occur. Because no two organic compounds (except enantiomers) have identical fingerprint patterns, matching this region against a computer database provides unambiguous identification of an unknown compound.	[1]
5	A	Option A is correct. The carbon–nitrogen triple bond (–C#N) produces a sharp, clean absorption band in the 2220–2260 cm ⁻¹ region, a unique spectral window where virtually no other functional groups absorb (except terminal alkynes C#C, which are much weaker).	[1]
6	D	Option D is correct. Primary amines have two N–H bonds that undergo coupled symmetric and asymmetric stretching vibrations, producing a diagnostic doublet (two peaks) in the 3300–3500 cm ⁻¹ region. Secondary amines have only one N–H bond and produce a single peak. Tertiary amines have no N–H bonds and show no peaks in this region.	[1]
7	B	Option B is correct. Alkenes are diagnosed by: (1) Weak-to-medium C=C stretching at 1620–1680 cm ⁻¹ ; (2) sp ² =C–H stretching occurring distinctly above 3000 cm ⁻¹ (3010–3095 cm ⁻¹), contrasting with saturated sp ³ C–H stretches which always occur below 3000 cm ⁻¹ (2850–2960 cm ⁻¹).	[1]
8	C	Option C is correct. For a vibration to absorb IR radiation, it must cause a change in dipole moment ($d_{\mu} / dr \neq 0$). In symmetrically substituted alkenes like trans-hex-3-ene (or ethene), the stretching of the non-polar C=C bond is completely symmetrical: dipole moment remains zero throughout the vibration, making the C=C stretch IR-inactive (forbidden). In unsymmetrical hex-1-ene, the vibration causes a dipole change and is IR-active.	[1]
9	B	Option B is correct. The formyl C–H bond of an aldehyde is weaker and more polarized than alkane C–H bonds. It exhibits two diagnostic absorptions in the 2720 cm ⁻¹ and 2820 cm ⁻¹ region due to Fermi resonance between the C–H fundamental stretch and an overtone of the C–H bending mode, uniquely distinguishing aldehydes from ketones.	[1]
10	A	Option A is correct. Propan-2-ol has an alcohol O–H group (3200–3600 cm ⁻¹). As oxidation proceeds to propanone (CH ₃ COCH ₃), the O–H group is destroyed and a ketone carbonyl (>C=O) is created. In the IR spectrum, the broad 3350 cm ⁻¹ band decreases in intensity and disappears, while a sharp, strong carbonyl absorption at 1715 cm ⁻¹ emerges.	[1]
11	D	Option D is correct. The reactants possess prominent O–H stretches: broad 2500–3300 cm ⁻¹ (acid) and 3200–3600 cm ⁻¹ (alcohol). In the pure ester ethyl ethanoate, all O–H groups are consumed. The IR spectrum exhibits exclusively C–H stretches (<3000 cm ⁻¹), an ester C=O stretch at 1735–1750 cm ⁻¹ , and strong C–O single-bond stretching bands at 1150–1250 cm ⁻¹ .	[1]
12	C	Option C is correct. Wavenumber ($\bar{\nu} = 1/\lambda$) represents the spatial frequency of the radiation in units of cm ⁻¹ . Because energy $E = h \cdot c \cdot \bar{\nu}$, wavenumber is directly proportional to both frequency and photon energy: higher wavenumbers correspond to higher vibrational energies and frequencies.	[1]
13	A	Option A is correct. Hooke's law for a vibrating spring models vibrational frequency as proportional to the square root of the bond force constant k divided by reduced mass μ : $\bar{\nu} = (1 / 2\pi \cdot c) \cdot \sqrt{k / \mu}$. A double bond (C=O) has a much higher force constant k (~1200 N/m) than a single bond (C–O, k ~500 N/m). Because the atoms involved are identical (C and O), the stiffer double bond vibrates at a significantly higher wavenumber.	[1]
14	C	Option C is correct. In Hooke's law, frequency is inversely proportional to the square root of the reduced mass: $\mu = (m_1 \cdot m_2) / (m_1 + m_2)$. Because hydrogen has an atomic mass of only 1, the reduced mass for C–H is ~0.92, whereas for C–Cl it is ~8.9. The light hydrogen atom oscillates much faster on the carbon spring than the heavy chlorine atom, giving a C–H stretch wavenumber (~3000 cm ⁻¹) more than four times higher than C–Cl (~700 cm ⁻¹).	[1]
15	B	Option B is correct. In conjugated carbonyls (C=C–C=O), resonance delocalization [C=C–C=O ↔ C ⁺ –C=C–O [–]] disperses pi electron density into the adjacent C–C bond. This reduces the bond order of the C=O group from ~2.0 to ~1.85. The weaker, less stiff carbonyl bond has a lower force constant k, causing the stretching absorption to shift to a lower wavenumber (~1685 cm ⁻¹ vs ~1730 cm ⁻¹).	[1]
16	D	Option D is correct. In liquid ethanol, each hydroxyl group participates in intermolecular hydrogen bonding (O–H ... O). The attraction of the neighboring oxygen pulls the hydrogen slightly away, weakening the covalent O–H bond (lower force constant, shifting frequency down to ~3350 cm ⁻¹). Because hydrogen bond lengths and angles vary dynamically in the liquid, a wide distribution of bond strengths exists, broadening the absorption band into a smooth curve.	[1]
17	C	Option C is correct. In an acid anhydride (R–CO–O–CO–R), the two carbonyl groups share a central oxygen atom. Their vibrations couple together, splitting into two distinct bands: an asymmetric stretch at ~1820 cm ⁻¹ and a symmetric stretch at ~1760 cm ⁻¹ . This diagnostic doublet in the 1750–1830 cm ⁻¹ zone uniquely identifies an anhydride.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
18	A	Option A is correct. Chlorine is highly electronegative (3.0 vs 2.5 for C) and exerts a powerful electron-withdrawing inductive effect (-I). By pulling electron density away from the carbonyl carbon, it forces the oxygen atom to share its electrons more tightly with carbon, increasing the C=O force constant and driving the stretching frequency up to ~1800 cm ⁻¹ (the highest among common carbonyl compounds).	[1]
19	D	Option D is correct. In amides, resonance delocalization of nitrogen's lone pair [R-C(=O)-NH ₂ <=> R-C(-O-)=N+H ₂] donates substantial electron density into the C-N bond while reducing the C=O bond order. This softens the carbonyl spring (lower force constant), shifting the carbonyl stretch down to 1640-1690 cm ⁻¹ .	[1]
20	B	Option B is correct. Aromatic rings exhibit three diagnostic regions: (1) sp ² C-H stretches at 3030-3100 cm ⁻¹ ; (2) sharp, medium-intensity skeletal C=C ring stretching bands around 1600 cm ⁻¹ and 1475-1500 cm ⁻¹ ; (3) intense C-H out-of-plane bending bands at 690-900 cm ⁻¹ that indicate the substitution pattern (e.g. mono-, ortho-, meta-, or para-substituted).	[1]
21	B	Option B is correct. Infrared absorption requires a changing dipole moment during vibration. Because N ₂ (N#N) and O ₂ (O=O) consist of identical atoms sharing electrons equally, their dipole moment is strictly zero at equilibrium and remains zero throughout any stretching vibration. Thus, they cannot couple with the electric field of an IR photon.	[1]
22	D	Option D is correct. CO ₂ already absorbs strongly across its primary absorption bands (e.g. 15 micrometers / 667 cm ⁻¹), causing near-saturation in those bands. Methane's C-H bending vibrations absorb intensely in the 7-8 micrometer (1250-1350 cm ⁻¹) region—an 'atmospheric window' where thermal radiation otherwise escapes directly to space—giving methane an immense radiative forcing per molecule.	[1]
23	A	Option A is correct. Water (H ₂ O) has a permanent dipole moment (1.85 D) and a bent geometry (104.5°). Its asymmetric stretch (3756 cm ⁻¹), symmetric stretch (3657 cm ⁻¹), and intense bending vibration (1595 cm ⁻¹)—broadened by extensive rotational transitions—absorb vast quantities of terrestrial IR radiation, accounting for ~60% of the natural greenhouse warming.	[1]
24	C	Option C is correct. Heteronuclear diatomic molecules (such as CO, NO, and HCl) possess permanent dipole moments due to electronegativity differences between the two dissimilar atoms. Stretching the bond modulates this dipole moment, creating an oscillating electric field that absorbs IR photons (CO absorbs at 2143 cm ⁻¹).	[1]
25	A	Option A is correct. Traditional transmission IR required grinding solids with nujol or KBr powder and pressing them into fragile transparent discs. ATR uses a high-refractive-index crystal (e.g. diamond or ZnSe). The sample is simply clamped directly onto the crystal surface; the IR beam undergoes total internal reflection, creating an evanescent wave that probes the sample surface within seconds with zero preparation.	[1]
26	D	Option D is correct. The Beer-Lambert law governs quantitative spectrophotometry: Absorbance A = log ₁₀ (I ₀ / I) = epsilon * c * l, where epsilon is the molar absorptivity, c is molar concentration, and l is the optical path length. A calibration plot of peak absorbance vs concentration is linear, allowing precise quantification of functional groups (e.g. ethanol in blood, plasticizers in polymers).	[1]
27	B	Option B is correct. Terminal alkynes (R-C#C-H) feature a characteristic, sharp sp C-H stretching band at ~3300 cm ⁻¹ (distinguishable from broad O-H bands) and an active C#C stretch at 2100-2140 cm ⁻¹ . Internal alkynes (R-C#C-R') lack the sp C-H bond (no 3300 cm ⁻¹ peak); furthermore, because the C#C bond is nearly symmetrical, its dipole change is minimal or zero, making the C#C stretch very weak or invisible.	[1]
28	C	Option C is correct. Fundamental IR bands correspond to vibrational transitions from v = 0 to v = 1. Due to molecular anharmonicity, transitions to v = 2 (first overtone) are weakly allowed, appearing at roughly double the fundamental frequency (~2 * nu ₀). For example, a strong C=O stretch at 1720 cm ⁻¹ often produces a faint overtone peak at ~3440 cm ⁻¹ .	[1]
29	B	Option B is correct. Polyatomic inorganic ions (like CO ₃ ²⁻ , NO ₃ ⁻ , SO ₄ ²⁻ , NH ₄ ⁺) have covalent bonds that absorb IR. The trigonal planar carbonate ion exhibits an intense, broad asymmetric C-O stretch at 1410-1450 cm ⁻¹ and an out-of-plane bend at 875-885 cm ⁻¹ (readily observed in limestone CaCO ₃ and calcium carbonate fillers). Monoatomic ions (Na ⁺ , Cl ⁻) have no covalent bonds and zero IR absorption.	[1]
30	A	Option A is correct. In a free carboxylic acid, the C=O bond is localized (1715 cm ⁻¹). In the carboxylate anion (R-COO ⁻), resonance delocalises the negative charge equally across both oxygen atoms, creating two equivalent C-O bonds with bond order 1.5. This resonance produces an intense asymmetric carboxylate stretch at 1550-1610 cm ⁻¹ and a symmetric stretch at ~1400 cm ⁻¹ , shifting the absorption down from the uncharged carbonyl position.	[1]
31	D	Option D is correct. Carbon-halogen single bonds (C-Cl, C-Br, C-I) involve heavy halogen atoms; according to Hooke's law, they vibrate at very low wavenumbers (C-Cl at 600-800 cm ⁻¹ , C-Br at 500-600 cm ⁻¹ , C-I < 500 cm ⁻¹). This low-frequency region is heavily cluttered by skeletal bending vibrations, and many standard optical components (like glass or quartz) absorb strongly here, making unambiguous identification difficult.	[1]
32	C	Option C is correct. Both are isomers of C ₂ H ₆ O. Isomer A contains an alcohol O-H group (broad band at 3350 cm ⁻¹), identifying ethanol. Isomer B lacks O-H and C=O bands, but shows a prominent C-O-C ether stretch at 1120 cm ⁻¹ , identifying methoxymethane (dimethyl ether).	[1]
33	A	Option A is correct. Aliphatic ketones exhibit carbonyl stretches around 1705-1725 cm ⁻¹ (centered at ~1715 cm ⁻¹). Esters exhibit carbonyl stretches at significantly higher wavenumbers (1735-1750 cm ⁻¹) due to the electron-withdrawing inductive effect (-I) of the adjacent electronegative ester oxygen atom, which increases the C=O force constant.	[1]
34	C	Option C is correct. The nitro group (-NO ₂) contains two equivalent, resonance-stabilized nitrogen-oxygen bonds. It exhibits two prominent diagnostic bands: an intense asymmetric N-O stretch at 1530-1560 cm ⁻¹ and a second strong symmetric N-O stretch at 1350-1380 cm ⁻¹ .	[1]
35	B	Option B is correct. The polar sulfur-oxygen coordinate/double bond produces intense IR absorptions: sulfoxides (R ₂ SO) show a single strong S=O stretch at 1030-1070 cm ⁻¹ ; sulfones (R ₂ SO ₂) exhibit two coupled vibrations: a symmetric stretch at 1120-1160 cm ⁻¹ and an asymmetric stretch at 1300-1350 cm ⁻¹ .	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
36	D	Option D is correct. Aromatic C-H bonds undergoing out-of-plane wagging (oop bending) interact strongly with adjacent ring hydrogens. The exact number of mutually adjacent hydrogens per ring (isolated, 2 adjacent, 3 adjacent, 4 adjacent) produces diagnostic frequencies between 700 and 900 cm^{-1} , enabling structural characterization of PAH isomers.	[1]
37	C	Option C is correct. A molecule with N atoms has 3N total degrees of freedom. For a non-linear molecule, 3 degrees represent translational motion and 3 represent rotational motion around the three Cartesian axes. Subtracting these gives $3N - 6$ fundamental vibrational degrees of freedom (for linear molecules, which have only 2 rotational degrees of freedom, the formula is $3N - 5$).	[1]
38	A	Option A is correct. For water, $N = 3$ and the molecule is bent (non-linear). Number of vibrational modes = $3N - 6 = (3 \times 3) - 6 = 3$. These three normal modes are: (1) Symmetric stretch (3657 cm^{-1}); (2) Asymmetric stretch (3756 cm^{-1}); (3) Symmetric bend / scissoring (1595 cm^{-1}). All three produce dipole changes and are IR-active.	[1]
39	D	Option D is correct. For a linear molecule, vibrational modes = $3N - 5 = (3 \times 3) - 5 = 4$. The modes are: (1) Symmetric stretch (1388 cm^{-1} , non-polar, zero dipole change, IR-inactive); (2) Asymmetric stretch (2349 cm^{-1} , dipole change, IR-active); (3 & 4) Two mutually perpendicular bending modes at 667 cm^{-1} of identical energy (degenerate, dipole change, IR-active). Thus, 3 modes absorb IR, making CO ₂ a powerful greenhouse gas.	[1]
40	B	Option B is correct. Degenerate vibrational modes have precisely the same force constant and geometry, resulting in identical energy levels and absorption frequencies. In linear CO ₂ , bending can occur in the horizontal plane (in-plane bend) or the vertical plane (out-of-plane bend). Because the cylindrical molecule is isotropic in both directions, both modes occur at precisely 667 cm^{-1} .	[1]
41	B	Option B is correct. Primary amides exhibit two sharp N-H spikes (symmetric and asymmetric stretching) in the 3180-3350 cm^{-1} region, contrasting sharply with the broad, ragged O-H envelope (2500-3300 cm^{-1}) of carboxylic acids. Furthermore, the amide carbonyl is shifted to a significantly lower wavenumber (1650-1680 cm^{-1}) than the carboxylic acid carbonyl (1715 cm^{-1}).	[1]
42	D	Option D is correct. Deuterium (2H, mass 2) is twice as heavy as protium (1H, mass 1). In Hooke's law, increasing the reduced mass μ by approximately a factor of two decreases the vibrational frequency by a factor of $1 / \sqrt{2} \sim 0.71$: $\nu(\text{O-D}) = \nu(\text{O-H}) / 1.37 = 3350 / 1.37 \sim 2450 \text{ cm}^{-1}$. This isotope shift confirms the identity of the vibrating bond.	[1]
43	A	Option A is correct. The primary natural greenhouse gases, H ₂ O and CO ₂ , have strong absorption bands below 8 micrometers (<1250 cm^{-1}) and above 14 micrometers (>700 cm^{-1}). The region between 8 and 14 micrometers is relatively transparent to terrestrial radiation, acting as an open window through which Earth radiates heat to space. Synthetic greenhouse gases (like CFCs and SF ₆) absorb directly in this window, causing disproportionate warming.	[1]
44	C	Option C is correct. CFCs and SF ₆ have C-F and S-F bonds with massive dipole moments, giving immense infrared absorption coefficients directly inside the unblocked atmospheric window (8-14 micrometers). Coupled with extraordinary atmospheric lifetimes (e.g. 3,200 years for SF ₆), a single molecule of SF ₆ has a warming potential over 23,000 times greater than a molecule of CO ₂ over 100 years.	[1]
45	A	Option A is correct. In the vapor phase at low pressure, molecules are widely separated and do not interact. The unhindered, non-hydrogen-bonded (free) O-H bond vibrates as an isolated oscillator, producing a sharp, narrow spike at 3680 cm^{-1} . In the liquid, dynamic hydrogen bonding broadens the absorption across 3200-3600 cm^{-1} and lowers the average frequency.	[1]
46	D	Option D is correct. In cyclopropanone, the interior C-C-C bond angle is compressed to 60° (requiring higher p-character in endocyclic bonds). Consequently, the exocyclic orbital directed toward the carbonyl oxygen acquires significantly higher s-character (approaching sp). Increased s-character shortens and strengthens the C=O bond, dramatically increasing its force constant and raising the stretching frequency to ~1815 cm^{-1} .	[1]
47	B	Option B is correct. Secondary alcohols oxidize to ketones: $\text{R-CH(OH)-R}' + [\text{O}] \rightarrow \text{R-CO-R}' + \text{H}_2\text{O}$. The alcohol O-H stretch (3200-3600 cm^{-1}) vanishes, and the characteristic ketone carbonyl stretch (1705-1725 cm^{-1}) appears.	[1]
48	C	Option C is correct. Poly(ethene) is a pure hydrocarbon showing only sp ³ C-H stretching (2850-2960 cm^{-1}) and -CH ₂ - bending (1465 cm^{-1} , 720 cm^{-1}). PVC contains polar C-Cl bonds, producing intense characteristic C-Cl stretching vibrations in the 600-750 cm^{-1} region that are completely absent in poly(ethene).	[1]
49	B	Option B is correct. Ordinary silicate glass and quartz contain Si-O covalent bonds that absorb intensely in the 1000 cm^{-1} region, rendering them completely opaque to mid-IR light. In contrast, ionic crystal lattices (NaCl, KBr) contain heavy ions (Na ⁺ , K ⁺ , Cl ⁻ , Br ⁻) held by non-directional electrostatic forces; their lattice vibrations occur far in the far-infrared (<400-600 cm^{-1}), making them optically transparent throughout the 4000-650 cm^{-1} region.	[1]
50	A	Option A is correct. Saturated aliphatic esters absorb at 1735-1750 cm^{-1} and acyl chlorides at ~1800 cm^{-1} . A lower carbonyl frequency of ~1690 cm^{-1} is diagnostic of conjugation (an alpha,beta-unsaturated ketone) or resonance donation from nitrogen (an amide).	[1]
51	D	Option D is correct. In the pure liquid state, carboxylic acids associate into cyclic dimers held together by two strong intermolecular hydrogen bonds: $[\text{R-C(=O...H-O)C-R}]$. These tight, symmetric hydrogen bonds dramatically perturb the O-H potential energy well, creating an intensely broad, jagged absorption envelope that extends from 2500 to 3300 cm^{-1} and completely engulfs the C-H stretches.	[1]
52	C	Option C is correct. Transmittance T is the fraction of incident light transmitted (I / I_0). Percentage transmittance %T = $(I / I_0) \times 100\%$. Absorbance $A = \log_{10}(I_0 / I) = -\log_{10}(T) = -\log_{10}(\%T / 100) = 2 - \log_{10}(\%T)$. When %T = 100%, A = 0; when %T = 10%, A = 1.0; when %T = 1%, A = 2.0.	[1]
53	A	Option A is correct. If solid particles in a KBr disc or mineral oil mull are larger than the wavelength of the incident IR light (2.5-15 micrometers), the particles scatter the beam rather than transmitting it. Because scattering intensity is inversely proportional to wavelength (Rayleigh scattering $\sim 1/\lambda^4$), shorter wavelengths (higher wavenumbers, 4000 cm^{-1}) scatter much more intensely, causing the baseline to slope downwards toward 4000 cm^{-1} .	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
54	C	Option C is correct. Propanoic acid is a carboxylic acid: it exhibits the hallmark broad O-H envelope (2500–3300 cm ⁻¹) and a carbonyl stretch at ~1715 cm ⁻¹ . Methyl ethanoate is an ester: it has zero O-H bonds (the spectrum is completely flat in the 2500–3600 cm ⁻¹ region except for sharp C-H peaks below 3000 cm ⁻¹) and its carbonyl stretch appears at a distinctly higher wavenumber (~1740 cm ⁻¹).	[1]
55	B	Option B is correct. Benzaldehyde possesses: (1) An aromatic phenyl ring (sp ² C-H >3000 cm ⁻¹ , sharp ring breathing at 1595 cm ⁻¹ and 1450 cm ⁻¹); (2) An aldehyde group (conjugated C=O at ~1700 cm ⁻¹ and the diagnostic C-H Fermi resonance doublet at 2730 and 2820 cm ⁻¹).	[1]
56	D	Option D is correct. The [M+1] peak arises due to the natural 1.1% abundance of carbon-13 (¹³ C). The number of carbon atoms, n, is calculated using the Cambridge formula: $n = (100 / 1.1) \times (\text{abundance of } [M+1] / \text{abundance of } M) = (100 / 1.1) \times (4.4 / 80.0) = 90.91 \times 0.0550 = 5.00$ carbon atoms. With 5 carbons (mass 60) and molecular mass 72, the formula is C ₅ H ₁₂ (pentane).	[1]
57	C	Option C is correct. Naturally occurring chlorine exists as two stable isotopes: ³⁵ Cl (~75%) and ³⁷ Cl (~25%), in a 3:1 ratio. A compound containing one chlorine atom exhibits a molecular ion peak [M] (containing ³⁵ Cl) and an [M+2] peak (containing ³⁷ Cl) with relative peak heights of exactly 3 : 1. (Bromine produces an [M] : [M+2] ratio of 1 : 1).	[1]
58	A	Option A is correct. Naturally occurring bromine exists as two stable isotopes of approximately equal abundance: ⁷⁹ Br (50.7%) and ⁸¹ Br (49.3%), yielding a characteristic 1:1 peak ratio for [M] and [M+2]. Bromoethane (C ₂ H ₅ Br) has molecular ion peaks at m/z = 29 + 79 = 108 and m/z = 29 + 81 = 110 of equal heights.	[1]
59	D	Option D is correct. Let the probability of ³⁵ Cl be p = 3/4 and ³⁷ Cl be q = 1/4. For two chlorine atoms, the isotope distribution is given by the binomial expansion (p + q) ² = p ² + 2pq + q ² : (3/4) ² = 9/16; 2 × (3/4) × (1/4) = 6/16; (1/4) ² = 1/16. Thus, the relative abundance ratio of [M] : [M+2] : [M+4] is strictly 9 : 6 : 1.	[1]
60	B	Option B is correct. For bromine, the probabilities of ⁷⁹ Br and ⁸¹ Br are approximately equal: p = 1/2 and q = 1/2. For two bromine atoms, the binomial expansion gives (p + q) ² = p ² + 2pq + q ² = (1/4) : (2/4) : (1/4) = 1 : 2 : 1. The three molecular ion peaks appear at intervals of 2 mass units in a 1 : 2 : 1 ratio.	[1]
61	B	Option B is correct. Probabilities: Cl (³⁵ Cl = 3/4, ³⁷ Cl = 1/4); Br (⁷⁹ Br = 1/2, ⁸¹ Br = 1/2). Expanding (3/4 + 1/4)(1/2 + 1/2): [M] (³⁵ Cl + ⁷⁹ Br) = 3/4 × 1/2 = 3/8; [M+2] (³⁵ Cl + ⁸¹ Br and ³⁷ Cl + ⁷⁹ Br) = (3/4 × 1/2) + (1/4 × 1/2) = 4/8; [M+4] (³⁷ Cl + ⁸¹ Br) = 1/4 × 1/2 = 1/8. Peak ratio is 3 : 4 : 1.	[1]
62	D	Option D is correct. Electron impact causes heterolytic/homolytic cleavage of the C-C bond in the molecular ion [CH ₃ -CH ₂ -CH ₃] ^{•+} → [CH ₃] ⁺ (m/z 15) + •CH ₂ CH ₃ (neutral radical, unobserved). The positively charged methyl cation [CH ₃] ⁺ is accelerated, deflected, and detected at m/z = 15.	[1]
63	A	Option A is correct. Propanal (Mr = 58) can fragment in two ways: (1) Cleavage of the C-C bond between ethyl and carbonyl yields the ethyl cation [C ₂ H ₅] ⁺ (mass 29) + •CHO; (2) The same cleavage can leave the positive charge on the formyl group: CH ₃ CH ₂ [•] + [CHO] ⁺ (mass 12 + 1 + 16 = 29). Both ions have m/z = 29.	[1]
64	C	Option C is correct. In methyl ketones (R-CO-CH ₃), alpha-cleavage adjacent to the carbonyl group readily loses the R [•] radical, producing the resonance-stabilized ethanoyl (acetyl) cation: [CH ₃ -C+=O ↔ CH ₃ -C#O ⁺]. With a mass of (12 × 2) + 3 + 16 = 43, this highly stable oxonium-type ion forms the base peak.	[1]
65	A	Option A is correct. 2-methylpropane ((CH ₃) ₃ CH, Mr = 58) loses a hydrogen radical (•H) with exceptional ease because the resulting tertiary carbocation [(CH ₃) ₃ C] ⁺ (m/z = 58 - 1 = 57) is stabilized by nine hyperconjugative C-H bonds and three electron-releasing (+I) methyl groups.	[1]
66	D	Option D is correct. Cleavage of the exocyclic bond connecting a substituent to an aromatic ring releases the phenyl cation [C ₆ H ₅] ⁺ : (6 × 12.0) + (5 × 1.0) = 77.0. A peak at m/z = 77 is a diagnostic indicator of a mono-substituted benzene ring.	[1]
67	B	Option B is correct. In alkylbenzenes (C ₆ H ₅ -CH ₂ -R), the benzylic C-C bond is exceptionally weak. Cleavage releases R [•] , generating the resonance-stabilized benzyl cation [C ₆ H ₅ CH ₂] ⁺ (m/z = 91). This ion spontaneously rearranges into the symmetrical, planar 6-pi-electron aromatic tropylium cation (cycloheptatrienyl cation), conferring extraordinary stability.	[1]
68	C	Option C is correct. Primary alcohols (R-CH ₂ -OH) undergo alpha-cleavage of the alkyl chain adjacent to the hydroxyl group: [R-CH ₂ -OH] ^{•+} → R [•] + [CH ₂ =O+H ↔ +CH ₂ -OH]. This resonance-stabilized oxonium ion has mass 12 + 2 + 16 + 1 = 31, providing a diagnostic fingerprint for primary alcohols.	[1]
69	B	Option B is correct. In propan-2-ol, the molecular ion [CH ₃ -CH(OH)-CH ₃] ^{•+} undergoes alpha-cleavage, expelling one of its methyl groups as a neutral radical: [CH ₃ -CH(OH)-CH ₃] ^{•+} → •CH ₃ (15) + [CH ₃ -CH=+OH] (m/z = 45). This secondary oxonium ion is resonance-stabilized and dominates the spectrum.	[1]
70	A	Option A is correct. Alcohols readily undergo thermal or electron-impact-induced elimination of water: [R-CH ₂ -CH ₂ -OH] ^{•+} → [R-CH=CH ₂] ^{•+} (m/z = M - 18) + H ₂ O (neutral, 18). This [M - 18] dehydration peak is a classic hallmark of alcohols.	[1]
71	D	Option D is correct. Formula: $n = (100 / 1.1) \times (4.3 / 65.0) = 90.91 \times 0.06615 = 6.01$ carbon atoms. With 6 carbons, mass from carbon = 6 × 12 = 72. Remaining mass = 86 - 72 = 14 hydrogens. Formula is C ₆ H ₁₄ (hexane).	[1]
72	C	Option C is correct. Formula: $n = (100 / 1.1) \times (\text{abundance } [M+1] / \text{abundance } M) = (100 / 1.1) \times (6.6 / 100) = 90.91 \times 0.066 = 6.0$ carbon atoms. With 6 carbons (mass 72) and total Mr = 78, the remaining 6 units are hydrogens, identifying benzene (C ₆ H ₆).	[1]
73	A	Option A is correct. Cleavage of the alkyl-carbonyl C-C bond in propanoic acid: [CH ₃ CH ₂ -COOH] ^{•+} → CH ₃ CH ₂ [•] (29) + [COOH] ⁺ (m/z = 12 + 32 + 1 = 45). The carboxyl cation [COOH] ⁺ is a characteristic marker of carboxylic acids.	[1]
74	C	Option C is correct. Loss of the neutral carboxyl radical (•COOH, mass 45) from the butanoic acid molecular ion leaves the propyl cation: [CH ₃ CH ₂ CH ₂ -COOH] ^{•+} → •COOH (45) + [CH ₃ CH ₂ CH ₂] ⁺ (m/z = 43).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
75	B	Option B is correct. In carbonyl compounds with an alkyl chain of at least three carbons (having gamma-hydrogens), the molecular ion undergoes a six-membered cyclic rearrangement: the carbonyl oxygen abstracts a gamma-H, triggering cleavage of the C _{alpha} - C _{beta} bond. This expels a neutral alkene (e.g. ethene, mass 28) and produces a stable enol radical cation (e.g. $[\text{CH}_3\text{C}(\text{OH})=\text{CH}_2]^+$, $m/z = 58$ for pentan-2-one).	[1]
76	D	Option D is correct. The molecular ion peak $[M]^+$ represents the intact molecule that has lost one electron; its m/z value equals the relative molecular mass M_r . The base peak is simply the most stable, abundant fragment ion formed during fragmentation; by convention, its intensity is normalized to 100%, and all other peaks are measured relative to it. The molecular ion may or may not be the base peak.	[1]
77	C	Option C is correct. Individual isotopes have exact atomic masses that deviate slightly from whole integers (mass defect: $^1\text{H} = 1.007825$, $^{12}\text{C} = 12.000000$, $^{16}\text{O} = 15.994915$). High-resolution instruments measure m/z to 0.0001 precision: propanone ($\text{C}_3\text{H}_6\text{O}$) has an exact mass of $(3 \times 12) + (6 \times 1.0078) + 15.9949 = 58.0417$, while butane (C_4H_{10}) is $(4 \times 12) + (10 \times 1.0078) = 58.0782$. This precise mass difference unambiguously distinguishes the two formulas.	[1]
78	A	Option A is correct. In the ionization chamber, an electrically heated filament emits a beam of electrons accelerated to 70 eV. When an energetic electron collides with a vaporized molecule, electrostatic repulsion ejects one electron from the molecule's highest occupied molecular orbital, creating a molecular radical cation: $M + e^- \rightarrow [M]^+ + 2e^-$.	[1]
79	D	Option D is correct. Ions accelerated by voltage V enter a magnetic field B . The magnetic Lorentz force provides centripetal acceleration: $B \cdot q \cdot v = (m \cdot v^2) / r$. Rearranging gives $r = (1/B) \cdot \sqrt{2 \cdot V \cdot (m/q)}$. For ions with charge +1 ($z = 1$), ions with lower mass (m) follow a path with a smaller radius (deflected more sharply), while heavier ions follow a wider path (deflected less). Varying B focuses ions of each m/z sequentially onto the detector.	[1]
80	B	Option B is correct. Mass spectrometers operate by manipulating charged particles using electromagnetic fields. Only positive ions ($[M]^+$, $[\text{fragment}]^+$) are accelerated by the electrostatic potential plates and deflected by the magnetic field onto the detector. Uncharged neutral radicals and molecules drift randomly and are continuously evacuated by high-vacuum pumps without generating an electrical signal.	[1]
81	B	Option B is correct. Because carbon and oxygen have even atomic masses (^{12}C , ^{16}O) and even valencies, and hydrogen has an odd mass (^1H) and odd valency (1), any stable neutral organic molecule containing only C, H, and O must have an EVEN molecular mass. Nitrogen has an even mass (^{14}N) but an ODD valency (3). Consequently, a neutral molecule with an odd number of nitrogen atoms always has an ODD nominal molecular mass (e.g. ethylamine $\text{C}_2\text{H}_7\text{N} = 45$).	[1]
82	D	Option D is correct. In neopentane, $\text{C}(\text{CH}_3)_4$, the central quaternary carbon is bonded to four methyl groups. Upon ionization, loss of one methyl radical ($^{\bullet}\text{CH}_3$) occurs with near-zero activation energy because it generates the immensely stable tert-butyl cation $[(\text{CH}_3)_3\text{C}]^+$ ($m/z = 57$). As a result, virtually 100% of the molecular ions fragment before reaching the detector, making the parent $[M]^+$ peak (m/z 72) imperceptibly tiny.	[1]
83	A	Option A is correct. Ethyl ethanoate cleaves on either side of the ester oxygen: (1) Cleavage of the C-O single bond: $[\text{CH}_3\text{CO}-\text{OCH}_2\text{CH}_3]^+ \rightarrow ^+\text{OCH}_2\text{CH}_3 + [\text{CH}_3\text{CO}]^+$ ($m/z = 43$, the base peak); (2) Cleavage leaving the charge on the alkyl group: $\text{CH}_3\text{COO}^+ + [\text{CH}_2\text{CH}_3]^+$ ($m/z = 29$).	[1]
84	C	Option C is correct. Using the binomial expansion $(3/4 + 1/4)^3 = (3/4)^3 + 3(3/4)^2(1/4) + 3(3/4)(1/4)^2 + (1/4)^3$: $[M] = 27/64$; $[M+2] = 27/64$; $[M+4] = 9/64$; $[M+6] = 1/64$. The relative peak height ratio is precisely 27 : 27 : 9 : 1.	[1]
85	A	Option A is correct. For three bromine atoms with $p = 1/2$ and $q = 1/2$: binomial expansion $(1/2 + 1/2)^3$ gives coefficients 1 : 3 : 3 : 1. The four peaks appear at m/z values separated by 2 units with relative intensities of 1 : 3 : 3 : 1.	[1]
86	D	Option D is correct. The 27 : 27 : 9 : 1 ratio across four peaks spaced by 2 mass units is the unique, definitive mathematical signature of THREE chlorine atoms ($(3/4 + 1/4)^3 = 27/64 : 27/64 : 9/64 : 1/64$).	[1]
87	B	Option B is correct. A mass loss of exactly 15 units ($\Delta m/z = 86 - 71 = 15$) represents the loss of a methyl radical ($^{\bullet}\text{CH}_3$, mass 12 + 3 = 15). In ketones, alpha-cleavage expels an alkyl radical to yield an acylium ion.	[1]
88	C	Option C is correct. 3-pentanone has two ethyl groups attached to the carbonyl. Symmetrical alpha-cleavage breaks one C-C bond, eliminating a neutral ethyl radical ($^{\bullet}\text{C}_2\text{H}_5$, mass 29): $[\text{CH}_3\text{CH}_2-\text{CO}-\text{CH}_2\text{CH}_3]^+ \rightarrow ^+\text{CH}_2\text{CH}_3$ (29) + $[\text{CH}_3\text{CH}_2\text{CO}]^+$ ($m/z = 57$).	[1]
89	B	Option B is correct. Molecular mass 56 corresponds to butene (C_4H_8 : $4 \times 12 + 8 = 56$). Cleavage of the allylic C-H or C-C bond in but-1-ene (loss of $^{\bullet}\text{CH}_3$, mass 15: $56 - 15 = 41$) produces the resonance-stabilized allyl carbocation $[\text{CH}_2=\text{CH}-\text{CH}_2^+ \leftrightarrow ^+\text{CH}_2-\text{CH}=\text{CH}_2]$, which produces an intense peak at $m/z = 41$.	[1]
90	A	Option A is correct. Sulfur consists of ^{32}S (~95.0%) and ^{34}S (~4.2%), separated by 2 mass units. A mono-sulfur compound exhibits an $[M+2]$ peak with a relative intensity of ~4.4% of the parent $[M]$ peak (much smaller than chlorine's 33% or bromine's 100%, but significantly larger than carbon/oxygen $[M+2]$ contributions).	[1]
91	D	Option D is correct. Both isomers have $M_r = 60$. In propan-1-ol, alpha-cleavage expels an ethyl radical ($^{\bullet}\text{CH}_2\text{CH}_3$, 29) to yield $[\text{CH}_2\text{OH}]^+$ at $m/z = 31$. In propan-2-ol, alpha-cleavage expels a methyl radical ($^{\bullet}\text{CH}_3$, 15) to yield $[\text{CH}_3\text{CHOH}]^+$ at $m/z = 45$. The contrasting base peaks (31 vs 45) distinguish the two constitutional isomers instantly.	[1]
92	C	Option C is correct. Pentan-2-one ($\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$) undergoes alpha-cleavage to produce the acetyl cation $[\text{CH}_3\text{CO}]^+$ (m/z 43) and has gamma-hydrogens that enable the McLafferty rearrangement to give $m/z = 58$. Symmetrical pentan-3-one ($\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$) cleaves on either side to yield exclusively the propanoyl cation $[\text{CH}_3\text{CH}_2\text{CO}]^+$ (m/z 57) and lacks gamma-hydrogens.	[1]
93	A	Option A is correct. Moles of C = $54.5 / 12 = 4.54$; H = $9.1 / 1 = 9.1$; O = $36.4 / 16 = 2.275$. Dividing by 2.275: C = 2.0, H = 4.0, O = 1.0. Empirical formula is $\text{C}_2\text{H}_4\text{O}$ (formula mass = 44.0). Ratio of M_r to empirical formula mass = $88 / 44 = 2$. Molecular formula is $(\text{C}_2\text{H}_4\text{O})_2 = \text{C}_4\text{H}_8\text{O}_2$.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
94	C	Option C is correct. The detector registers an electrical current when charged ions strike its surface. The ethyl cation $[\text{CH}_3\text{CH}_2]^+$ carries a positive charge and is focused by the electric and magnetic fields onto the detector (producing a peak at $m/z = 29$). The chlorine radical ($^{\bullet}\text{Cl}$) has no charge; it cannot be accelerated or steered by electric/magnetic fields and is pumped out by the vacuum system.	[1]
95	B	Option B is correct. Electron Impact (70 eV) imparts large excess internal energy, shattering molecules into dozens of fragments and often destroying the parent molecular ion. Electrospray Ionization (ESI) is a 'soft' technique: it gently transfers pre-existing ions from an aerosol solution into the gas phase by protonation ($[\text{M} + \text{H}]^+$), keeping fragile macromolecules (proteins, DNA, polymers) intact for molecular mass determination.	[1]
96	D	Option D is correct. Carbon in nature consists of 98.9% ^{12}C and 1.1% ^{13}C . For a molecule with n carbon atoms, the probability of containing exactly one ^{13}C atom is approximately $n \times 1.1\%$. Therefore, the ratio of $[\text{M}+1]$ abundance to $[\text{M}]$ abundance is $(n \times 1.1) / 100$. Rearranging for n yields $n = (100 / 1.1) \times (\text{abundance } [\text{M}+1] / \text{abundance } \text{M})$.	[1]
97	C	Option C is correct. If an ion $[\text{m}_1]^+$ fragments into $[\text{m}_2]^+$ after acceleration but before entering the magnetic analyzer, the daughter ion retains the velocity of the parent but has the lower mass of the daughter. The magnetic field deflects it according to an apparent mass $m^* = (m_2)^2 / m_1$, producing a characteristically broad, low-intensity, non-integer peak that confirms the precursor-daughter fragmentation pathway.	[1]
98	A	Option A is correct. The IR peak at 1715 cm^{-1} confirms a carbonyl group ($>\text{C}=\text{O}$). The lack of peaks $>3000 \text{ cm}^{-1}$ and lack of aldehyde C-H doublet rules out alcohols, carboxylic acids, and aldehydes. Mass 58 with base peak at $m/z = 43$ ($[\text{CH}_3\text{CO}]^+$) uniquely identifies the symmetrical ketone propanone (CH_3COCH_3 : $15 + 28 + 15 = 58$).	[1]
99	D	Option D is correct. Equal peak heights at $m/z = 122$ and 124 ($[\text{M}] : [\text{M}+2] = 1 : 1$) confirm the presence of ONE bromine atom. Subtracting the mass of bromine: $122 - 79 = 43$. A hydrocarbon fragment of mass 43 corresponds to the propyl group, C_3H_7 ($3 \times 12 + 7 = 43$), identifying bromopropane ($\text{C}_3\text{H}_7\text{Br}$).	[1]
100	B	Option B is correct. In MS/MS, two mass spectrometers are coupled in series with a collision cell between them. MS-1 acts as a mass filter, isolating a specific ion from a complex biological mixture (e.g. blood serum). In the collision cell, the ion collides with inert gas (argon/helium), shattering into fragments. MS-2 then records the fragmentation spectrum, providing definitive sequence and structural information.	[1]
101	C	Option C is correct. Using the standard Cambridge formula for carbon-13 abundance: $n = (100 / 1.1) \times (\text{abundance of } [\text{M}+1] / \text{abundance of } \text{M}) = (100 / 1.1) \times (2.2 / 50.0) = 90.91 \times 0.0440 = 4.00$ carbon atoms.	[1]
102	A	Option A is correct. Naturally occurring chlorine consists of ^{35}Cl and ^{37}Cl in approximately a 3 : 1 ratio (75.8% to 24.2%), producing an $[\text{M}] : [\text{M}+2]$ ratio of ~3 : 1. Naturally occurring bromine consists of ^{79}Br and ^{81}Br in virtually equal proportions (50.7% to 49.3%), producing an $[\text{M}] : [\text{M}+2]$ ratio of ~1 : 1.	[1]
103	D	Option D is correct. The O-H stretch of an alcohol is a smooth, broad band centered at 3350 cm^{-1} ($3200\text{--}3600 \text{ cm}^{-1}$) with no carbonyl absorption. The O-H stretch of a carboxylic acid is exceptionally broad and jagged, extending from 2500 to 3300 cm^{-1} and overlapping the aliphatic C-H stretches at $\sim 2950 \text{ cm}^{-1}$, accompanied by an intense carbonyl ($\text{C}=\text{O}$) stretch at $1700\text{--}1725 \text{ cm}^{-1}$.	[1]
104	B	Option B is correct. In an ester ($\text{R}-\text{CO}-\text{O}-\text{R}'$), the highly electronegative ester oxygen pulls electron density away from the carbonyl carbon via an inductive electron-withdrawing effect ($-I$). This polarization causes the carbonyl carbon to pull electrons tighter from the carbonyl oxygen, shortening and stiffening the $\text{C}=\text{O}$ double bond (higher force constant k), which shifts the stretching frequency to a higher wavenumber ($1735\text{--}1750 \text{ cm}^{-1}$ vs $1705\text{--}1725 \text{ cm}^{-1}$ in ketones).	[1]
105	A	Option A is correct. The carbon-nitrogen triple bond ($-\text{C}\equiv\text{N}$) produces a sharp, clean absorption band in the $2220\text{--}2260 \text{ cm}^{-1}$ region. No other common functional group absorbs in this unique spectral window (except terminal alkynes, which are much weaker).	[1]
106	C	Option C is correct. Chlorine isotopes have probabilities $^{35}\text{Cl} = 3/4$ and $^{37}\text{Cl} = 1/4$. For two chlorine atoms, expanding $(3/4 + 1/4)^2$ yields: $[\text{M}] = (3/4)^2 = 9/16$; $[\text{M}+2] = 2 \times (3/4) \times (1/4) = 6/16$; $[\text{M}+4] = (1/4)^2 = 1/16$. The relative ratio is 9 : 6 : 1.	[1]
107	B	Option B is correct. The molecular ion $[\text{CH}_3\text{COCH}_3]^+$ undergoes alpha-cleavage adjacent to the carbonyl group, losing a methyl radical ($^{\bullet}\text{CH}_3$, mass 15) to form the resonance-stabilized ethanoyl (acylium) cation: $[\text{CH}_3-\text{C}^+=\text{O} \leftrightarrow \text{CH}_3-\text{C}\equiv\text{O}^+]$. Its mass is $(12 \times 2) + (3 \times 1) + 16 = 43$.	[1]
108	D	Option D is correct. For a molecular vibration to absorb an infrared photon, the oscillating motion of the atoms must create an oscillating dipole moment ($d_{\text{mu}} / dr \neq 0$) that can interact with the alternating electric field of the infrared electromagnetic wave. Symmetrical vibrations that maintain zero dipole moment (like in N_2 , O_2 , or symmetric CO_2 stretch) are IR-inactive.	[1]
109	A	Option A is correct. A primary amine has two N-H bonds attached to the same nitrogen atom. Coupled stretching gives rise to two distinct normal modes: symmetric stretching and asymmetric stretching, creating a diagnostic doublet (two peaks) in the $3300\text{--}3500 \text{ cm}^{-1}$ region. A secondary amine has only one N-H bond and therefore exhibits only a single N-H stretching peak.	[1]
110	C	Option C is correct. Data synthesis: (1) Mass 60 with 3 carbons indicates $\text{C}_3\text{H}_8\text{O}$ ($(3 \times 12) + 8 + 16 = 60$); (2) IR smooth broad band at 3350 cm^{-1} with no $\text{C}=\text{O}$ stretch confirms an alcohol (ruling out ethanoic acid and methoxymethane); (3) Mass spectrum base peak at $m/z = 31$ is the hydroxymethyl cation $[\text{CH}_2\text{OH}]^+$, which is the diagnostic alpha-cleavage fragment of a PRIMARY alcohol (loss of ethyl radical C_2H_5 , 29 : $60 - 29 = 31$). Propan-2-ol would give a base peak at $m/z = 45$ ($[\text{CH}_3\text{CHOH}]^+$). Therefore, the compound is definitively propan-1-ol.	[1]